

Supplement to: Where Does Jagged Competence Come From?

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This supplement retains analytical condition comparisons, controls, settings and support. Exhaustive records are released separately as `paper/data/release/*.json.gz`. Each table names its stable detailed file; the schema and SHA-256 identities are in `release/README.md` and `release/index.json`. Ranges across cells are descriptive, not pooled estimates or confidence intervals. Table numbers are fixed by the build script. This is part of the proposed evidence release, not an independent experiment. Code: MIT; paper and released data: CC BY 4.0.

Table S1: Five achievements: complete scoped condition matrix. Original-law conditions; three seeds each. Supported is limited to the named measurement and support, not supported means a tested criterion failed, and not established means it was not established by the available measure. Mixed pass counts retain failing seeds. Frozen preservation is by verified design, not evidence that further training would preserve a live computation. Probe recovery is not use. Computation marks refer only to the audited sums output; failure does not establish that no internal computation exists. Board-output counts refer to rendering 0; other renderings are reported separately in Supplement Table S16. Local erosion uses 1,555 histories, not a board census. Full per-seed counts, controls and ceilings follow in this supplement. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: `paper/data/release/table1_full.json.gz` (schema and SHA-256 in `release/index.json`).

condition	recoverability	computation	access	preservation	reliable prediction
r8: records+sums → prediction (sums route only)	not established	supported (frozen)	supported by design (exact sums supplied)	supported (frozen)	supported 2/3
r8: records+sums → prediction	not established	supported (frozen)	supported by design (exact sums supplied)	supported (frozen)	supported 3/3
r8: records → prediction	slot 1; raw linear	not established	not established	not established	not supported 0/3
r10: records+sums → prediction (sums route only)	supported slots: 0/5 / 0/5 / 0/5	supported (frozen)	supported by design (exact sums supplied)	supported (frozen)	not supported 0/3
r10: records → prediction+sums	supported slots: 5/5 / 5/5 / 5/5	not established	not established	not established	not supported 0/3
r10: records → prediction	supported slots: 5/5 / 5/5 / 5/5	not established	not established	not established	not supported 0/3
r11: records → prediction+sums (sums dose 100%)	supported slots: 5/5 / 5/5 / 5/5	not supported 0/3	not established	not established	not supported 0/3
r11: records → prediction+sums (sums dose 10%)	supported slots: 5/5 / 5/5 / 5/5	not supported 0/3	not established	not established	not supported 0/3
r11: records → prediction+sums (sums dose 1%)	supported slots: 5/5 / 5/5 / 5/5	not supported 0/3	not established	not established	not supported 0/3

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condition	recoverability	computation	access	preservation	reliable prediction
r11: records → prediction+sums (Erosion continuation)	supported slots: 3/5 / 1/5 / 1/5	supported 1/3	not established	not supported (inexact)	supported 1/3
r11: records → prediction (Erosion continuation)	supported slots: 1/5 / 1/5 / 1/5	not supported 0/3	not established	not supported (inexact)	not supported 0/3
r11: records → prediction (cutoff coverage)	supported slots: 1/5 / 1/5 / 1/5	not supported 0/3	not established	not established	not supported 0/3
r11: records → prediction (decoy coverage)	supported slots: 1/5 / 1/5 / 1/5	not supported 0/3	not established	not established	not supported 0/3
r11: records → prediction (uniform sampling)	supported slots: 1/5 / 1/5 / 1/5	not supported 0/3	not established	not established	not supported 0/3
r12: records+sums → prediction	supported slots: 2/5 / 2/5 / 3/5	supported 3/3	supported by design (exact sums supplied)	supported (frozen)	supported 2/3
r12: records → prediction+sums	supported slots: 5/5 / 5/5 / 5/5	not supported 0/3	not established	not established	not supported 0/3
r12: records → prediction (cutoff coverage)	supported slots: 1/5 / 1/5 / 1/5	not supported 0/3	not established	not established	not supported 0/3
r12: records → prediction (decoy coverage)	supported slots: 1/5 / 1/5 / 1/5	not supported 0/3	not established	not established	not supported 0/3
r12: records → prediction (uniform sampling)	supported slots: 1/5 / 1/5 / 1/5	not supported 0/3	not established	not established	not supported 0/3
r13: records → prediction+sums	not established	not supported 0/3	not established	not established	not supported 0/3
r13: records+sums → prediction (exact supplied)	not established	supported 3/3	supported by design (exact sums supplied)	supported (frozen)	supported 1/3
r13: Connected (frozen)	not established	not supported 0/3	demonstrated by matched connection intervention (not selective use)	supported (frozen)	not supported 0/3
r13: Connected (live)	not established	not supported 0/3	demonstrated by matched connection intervention (not selective use)	not established	not supported 0/3
r13: records+sums → prediction (Numerical encoding)	not established	supported 3/3	supported by design (exact sums supplied)	supported (frozen)	not supported 0/3
r13: AdamW	not established	not supported	not established	not supported	not established
r13: blocked prediction gradient	not established	supported	not established	supported	not established
r13: reduced-lr AdamW	not established	not supported	not established	not supported	not established
r13: SGD	not established	not supported	not established	not supported	not established
r13: records → prediction (sampled targets)	not established	not supported 0/3	not established	not established	not supported 0/3
r13: records → prediction+sums (Probability targets)	not established	not supported 0/3	not established	not established	not supported 0/3
r13: records → prediction (Probability targets)	not established	not supported 0/3	not established	not established	not supported 0/3
r13: records → prediction+sums (sampled targets)	not established	not supported 0/3	not established	not established	not supported 0/3

Table S2: Access and supplied sums: complete per-seed comparison. Every seed is explicit; compare access variants within r13 and supplied versus target sums within r12. Across studies these are historical, not matched comparisons; r10 is a different all-slots task. Counts use rendering 0 where recorded; missing census counts are not zero. *r8 reports full-law maximum, r10 law-row maximum, r12/r13 N3–N8 maximum. Original-task history TV bar is 0.02; r10 original-law law-row bar is 0.05, equal-law 0.02. Finite KL/TV use five significant digits; counts are exact. A pass is not correctness everywhere. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/table2_full.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

study / condition / seed	B pass	natural KL (bits)	misreads (boards)	short TV max	long TV max*	endpoint label
r8 records+sums → prediction (sums route only) / 0	yes	0.00033865	not recorded	not recorded	0.017721	B_PASS
r8 records+sums → prediction (sums route only) / 1	yes	6.7874e-05	not recorded	not recorded	0.011403	B_PASS
r8 records+sums → prediction (sums route only) / 2	no	6.5306e-05	not recorded	not recorded	0.020775	B_INCOMPLETE
r8 records+sums → prediction / 0	yes	0.00025483	not recorded	not recorded	0.018665	B_PASS
r8 records+sums → prediction / 1	yes	6.8052e-05	not recorded	not recorded	0.0051108	B_PASS
r8 records+sums → prediction / 2	yes	8.0944e-05	not recorded	not recorded	0.0059933	B_PASS
r8 records → prediction / 0	no	0.00029829	not recorded	not recorded	0.25	B_INCOMPLETE
r8 records → prediction / 1	no	7.047e-05	not recorded	not recorded	0.2548	B_INCOMPLETE
r8 records → prediction / 2	no	0.00021024	not recorded	not recorded	0.25649	B_INCOMPLETE
r10 records+sums → prediction (sums route only) / 0	no	0.0070603	not recorded	not recorded	0.12242	INCOMPLETE
r10 records+sums → prediction (sums route only) / 1	no	0.0059933	not recorded	not recorded	0.079414	INCOMPLETE
r10 records+sums → prediction (sums route only) / 2	no	0.0057117	not recorded	not recorded	0.10517	INCOMPLETE
r10 records → prediction / 0	no	0.016034	not recorded	not recorded	0.3219	INCOMPLETE
r10 records → prediction / 1	no	0.019138	not recorded	not recorded	0.28963	INCOMPLETE
r10 records → prediction / 2	no	0.016491	not recorded	not recorded	0.30706	INCOMPLETE
r10 records → prediction+sums / 0	no	0.032325	not recorded	not recorded	0.27737	INCOMPLETE
r10 records → prediction+sums / 1	no	0.033959	not recorded	not recorded	0.25519	INCOMPLETE
r10 records → prediction+sums / 2	no	0.035928	not recorded	not recorded	0.23295	INCOMPLETE
r12 records+sums → prediction / 0	yes	0.00011363	0	0.015676	0.014557	PASS
r12 records+sums → prediction / 1	yes	0.00029505	1	0.0093861	0.010587	PASS
r12 records+sums → prediction / 2	no	0.00025691	1	0.01258	0.012218	B_INCOMPLETE
r12 records → prediction+sums / 0	no	0.0015898	473	0.049233	0.064965	B_INCOMPLETE
r12 records → prediction+sums / 1	no	0.0020044	648	0.13295	0.087776	B_INCOMPLETE
r12 records → prediction+sums / 2	no	0.0022981	815	0.03955	0.044799	B_INCOMPLETE
r13 records → prediction+sums / 0	no	0.0021068	331	0.081309	0.0985	INCOMPLETE

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study / condition / seed	B pass	natural KL (bits)	misreads (boards)	short TV max	long TV max*	endpoint label
r13 records → prediction+sums / 1	no	0.0018252	725	0.044418	0.10016	INCOMPLETE
r13 records → prediction+sums / 2	no	0.0020459	637	0.047182	0.072752	INCOMPLETE
r13 records+sums → prediction (exact supplied) / 0	no	0.00011384	0	0.022749	0.013305	INCOMPLETE
r13 records+sums → prediction (exact supplied) / 1	no	0.00030667	0	0.012822	0.01569	INCOMPLETE
r13 records+sums → prediction (exact supplied) / 2	yes	0.00028061	2	0.011454	0.010661	PASS
r13 Connected (frozen) / 0	no	0.00062178	259	0.013796	0.24257	INCOMPLETE
r13 Connected (frozen) / 1	no	0.0005485	69	0.014747	0.011304	INCOMPLETE
r13 Connected (frozen) / 2	no	0.00034049	58	0.013029	0.015699	INCOMPLETE
r13 Connected (live) / 0	no	0.00031731	117	0.025362	0.24683	INCOMPLETE
r13 Connected (live) / 1	no	0.00041094	43	0.015556	0.25128	INCOMPLETE
r13 Connected (live) / 2	no	0.00029062	39	0.01333	0.011997	INCOMPLETE
r13 records+sums → prediction (Numerical encoding) / 0	no	0.00083668	317	0.028812	0.24067	INCOMPLETE
r13 records+sums → prediction (Numerical encoding) / 1	no	0.00079477	121	0.042917	0.069823	INCOMPLETE
r13 records+sums → prediction (Numerical encoding) / 2	no	0.00054318	10	0.021251	0.18902	INCOMPLETE
r8 board-only control	no	0.013514	not recorded	not recorded	0.1875	null, calibration
exact task oracle	yes	0	0	0	0	ceiling, not trained

Table S3: Registered studies, corrected audits and supplementary analyses. r8/r9 are CPU-era; later GPU execution metadata are kept distinct. Counts inventory manifest authorities, not independent experiments. Exact historical commands/hardware absent from run summaries remain unknown; instruction files are not proof of execution. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_A_history.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

study	recorded hardware	version/ review records	historical scope
r8	CPU-era; processor model not recorded in saved per-run summaries	11	Registered endpoints retained; corrected audits and supplementary analyses are separate from training authority. Command instructions alone do not establish the hardware or command actually used.
r9	CPU-era; processor model not recorded in saved per-run summaries	1	Registered endpoints retained; corrected audits and supplementary analyses are separate from training authority. Command instructions alone do not establish the hardware or command actually used.
r10	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	20	Registered endpoints retained; corrected audits and supplementary analyses are separate from training authority. Command instructions alone do not establish the hardware or command actually used.

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study	recorded hardware	version/ review records	historical scope
r11	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	13	Registered endpoints retained; corrected audits and supplementary analyses are separate from training authority. Command instructions alone do not establish the hardware or command actually used.
r12	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	14	Registered endpoints retained; corrected audits and supplementary analyses are separate from training authority. Command instructions alone do not establish the hardware or command actually used.
r13	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	44	Registered endpoints retained; corrected audits and supplementary analyses are separate from training authority. Command instructions alone do not establish the hardware or command actually used.

Table S4: r8 sampling, training and registered criteria. Read directly from the hash-verified registration (r9: read-only scope). This table retains declared sampling, truncation, coverage, loss, settings and criteria, not inferred hardware. Long verbatim readings, r10’s full rational conditional-frequency table and source identities remain in the manifest-bound registration; r10 is a different task. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/prepared_r8.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

registered property	value
criteria/KL_bits	0.01
criteria/law_and_intervention_TV	0.02
criteria/witness_recovery	0.8
linear_probe_settings/C	10
linear_probe_settings/class_weight	balanced
linear_probe_settings/estimator	LogisticRegression
linear_probe_settings/max_iter	5,000
linear_probe_settings/ standardization	StandardScaler_fit_on_probe_training_only
linear_probe_settings/tol	1e-05
settings/audit_steps	[0, 1000, 2000, 5000, 10000, 15000, 20000]
settings/batch	256
settings/futility_at	5,000
settings/futility_gap_fraction	0.1
settings/laws	["original", "equal"]
settings/lr	0.003
settings/routes	["a_only", "raw_only", "dual"]
settings/seeds	[0, 1, 2]
settings/steps	20,000
settings/weight_decay	0.01

Table S5: r8 complete endpoint version matrix. All recorded conditions and seeds, including equal-law controls; no endpoint is selected. Equal-law category signatures are undefined, not zero. Five significant digits for continuous values. *Full-law / long / law-row support follows Table 2. Exact-oracle errors are zero; matched initial controls and complete deciding-function calibration are in Appendix C. Census counts are rendering 0; all four renderings and saved-r9 counts are in the census table. r11 reports B_law.maximum_TV on all history cases. Missing measurements stay not recorded. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/appendix_B_r8.json.gz (schema and SHA-256 in <release/index.json>).

condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records+sums → prediction (sums route only)	equal	0	yes	9.3987e-05	not recorded	not recorded	not recorded	B_PASS
records+sums → prediction (sums route only)	equal	1	yes	3.748e-06	not recorded	not recorded	not recorded	B_PASS
records+sums → prediction (sums route only)	equal	2	yes	2.9542e-05	not recorded	not recorded	not recorded	B_PASS
records+sums → prediction (sums route only)	original	0	yes	0.00033865	not recorded	not recorded	0.017721	B_PASS
records+sums → prediction (sums route only)	original	1	yes	6.7874e-05	not recorded	not recorded	0.011403	B_PASS
records+sums → prediction (sums route only)	original	2	no	6.5306e-05	not recorded	not recorded	0.020775	B_INCOMPLETE
records+sums → prediction	equal	0	yes	9.3783e-05	not recorded	not recorded	not recorded	B_PASS
records+sums → prediction	equal	1	yes	3.7485e-06	not recorded	not recorded	not recorded	B_PASS
records+sums → prediction	equal	2	yes	2.9543e-05	not recorded	not recorded	not recorded	B_PASS
records+sums → prediction	original	0	yes	0.00025483	not recorded	not recorded	0.018665	B_PASS
records+sums → prediction	original	1	yes	6.8052e-05	not recorded	not recorded	0.0051108	B_PASS
records+sums → prediction	original	2	yes	8.0944e-05	not recorded	not recorded	0.0059933	B_PASS
records → prediction	equal	0	yes	9.227e-05	not recorded	not recorded	not recorded	B_PASS
records → prediction	equal	1	yes	4.2927e-06	not recorded	not recorded	not recorded	B_PASS
records → prediction	equal	2	yes	3.009e-05	not recorded	not recorded	not recorded	B_PASS
records → prediction	original	0	no	0.00029829	not recorded	not recorded	0.25	B_INCOMPLETE
records → prediction	original	1	no	7.047e-05	not recorded	not recorded	0.2548	B_INCOMPLETE
records → prediction	original	2	no	0.00021024	not recorded	not recorded	0.25649	B_INCOMPLETE

Table S6: r9 sampling, training and registered criteria. Read directly from the hash-verified registration (r9: read-only scope). This table retains declared sampling, truncation, coverage, loss, settings and criteria, not inferred hardware. Long verbatim readings, r10’s full rational conditional-frequency table and source identities remain in the manifest-bound registration; r10 is a different task. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/prepared_r9.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

registered property	value
type	Read-only diagnosis of saved r8 networks; no updates
scope	Completed boards, clear-neutral runs and failing saved histories; per-seed support is retained below

Table S7: r9 complete endpoint version matrix. All recorded conditions and seeds, including equal-law controls; no endpoint is selected. Equal-law category signatures are undefined, not zero. Five significant digits for continuous values. *Full-law / long / law-row support follows Table 2. Exact-oracle errors are zero; matched initial controls and complete deciding-function calibration are in Appendix C. Census counts are rendering 0; all four renderings and saved-r9 counts are in the census table. r11 reports B_law.maximum_TV on all history cases. Missing measurements stay not recorded. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_B_r9.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records+sums → prediction (sums route only)	original	0	not recorded	0.00033865	0	1	not recorded	not assessed in r9
records+sums → prediction (sums route only)	original	1	not recorded	6.7874e-05	0	1	not recorded	not assessed in r9
records+sums → prediction (sums route only)	original	2	not recorded	6.5306e-05	0	34	not recorded	not assessed in r9
records+sums → prediction	original	0	not recorded	0.00025483	0	0	not recorded	not assessed in r9
records+sums → prediction	original	1	not recorded	6.8052e-05	0	0	not recorded	not assessed in r9
records+sums → prediction	original	2	not recorded	8.0944e-05	0	0	not recorded	not assessed in r9
records → prediction	original	0	not recorded	0.00029829	8	69	not recorded	not assessed in r9
records → prediction	original	1	not recorded	7.047e-05	4	25	not recorded	not assessed in r9
records → prediction	original	2	not recorded	0.00021024	3	66	not recorded	not assessed in r9

Table S8: r10 sampling, training and registered criteria. Read directly from the hash-verified registration (r9: read-only scope). This table retains declared sampling, truncation, coverage, loss, settings and criteria, not inferred hardware. Long verbatim readings, r10’s full rational conditional-frequency table and source identities remain in the manifest-bound registration; r10 is a different task. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/prepared_r10.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

registered property	value
criteria/constructed_maximum_TV	0.05
criteria/donor_follow	0.8
criteria/equal_law_TV	0.02

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registered property	value
criteria/futility_gap_fraction	0.1
criteria/minimum_accurate__	20
interchange_pairs	
criteria/minimum_evaluation_cell__	20
observations	
criteria/natural_and_covered_KL__	0.1
fraction_of_blind	
criteria/null_donor_follow	0.2
criteria/positive_donor_follow	0.95
criteria/probe_positive_accuracy	0.99
settings/CPU_threads_per_process	1
settings/MLP_fit_device	audit_device
settings/arms	["free", "free_a", "a_forced"]
settings/audits	[0, 1000, 2000, 5000, 10000, 15000, 20000]
settings/batch	256
settings/device_default	cuda
settings/execution_widths/a_forced	1
settings/execution_widths/free	3
settings/execution_widths/free_a	3
settings/laws	["original", "equal"]
settings/linear_fit_device	cpu
settings/linear_settings/C	10
settings/linear_settings/class_weight	balanced
settings/linear_settings/max_iter	5,000
settings/linear_settings/	training_only
standardization	
settings/linear_settings/tol	1e-05
settings/lr	0.003
settings/maximum_processes	5
settings/precision	float32
settings/probe_seed	2,026,101,003
settings/rounds	8
settings/seeds	[0, 1, 2]
settings/steps	20,000
settings/stream_seed_base	2,026,101,001
settings/test_rounds	12
settings/test_seed	2,026,101,002
settings/weight_decay	0.01

Table S9: r10 complete endpoint version matrix. All recorded conditions and seeds, including equal-law controls; no endpoint is selected. Equal-law category signatures are undefined, not zero. Five significant digits for continuous values. *Full-law / long / law-row support follows Table 2. Exact-oracle errors are zero; matched initial controls and complete deciding-function calibration are in Appendix C. Census counts are rendering 0; all four renderings and saved-r9 counts are in the census table. r11 reports B_{law}.maximum_TV on all history cases. Missing measurements stay not recorded. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_B_r10.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records+sums → prediction (sums route only)	equal	0	yes	5.8034e-05	not recorded	not recorded	0.0041717	PASS
records+sums → prediction (sums route only)	equal	1	yes	1.2963e-05	not recorded	not recorded	0.0019756	PASS
records+sums → prediction (sums route only)	equal	2	yes	3.9156e-05	not recorded	not recorded	0.0032849	PASS

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condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records+sums → prediction (sums route only)	original	0	no	0.0070603	not recorded	not recorded	0.12242	constructed_law_TV, same_board_witness
records+sums → prediction (sums route only)	original	1	no	0.0059933	not recorded	not recorded	0.079414	constructed_law_TV, same_board_witness
records+sums → prediction (sums route only)	original	2	no	0.0057117	not recorded	not recorded	0.10517	constructed_law_TV, same_board_witness
records → prediction+sums	equal	0	yes	6.4335e-05	not recorded	not recorded	0.005519	PASS
records → prediction+sums	equal	1	yes	1.6126e-05	not recorded	not recorded	0.0020335	PASS
records → prediction+sums	equal	2	yes	4.147e-05	not recorded	not recorded	0.0040652	PASS
records → prediction+sums	original	0	no	0.032325	not recorded	not recorded	0.27737	constructed_law_TV, covered_value_KL, natural_KL, same_board_witness
records → prediction+sums	original	1	no	0.033959	not recorded	not recorded	0.25519	constructed_law_TV, covered_value_KL, natural_KL, same_board_witness
records → prediction+sums	original	2	no	0.035928	not recorded	not recorded	0.23295	constructed_law_TV, covered_value_KL, natural_KL, same_board_witness
records → prediction	equal	0	yes	6.4992e-05	not recorded	not recorded	0.0046541	PASS
records → prediction	equal	1	no	0.00088019	not recorded	not recorded	0.042784	constructed_law_TV
records → prediction	equal	2	yes	3.9182e-05	not recorded	not recorded	0.0032866	PASS
records → prediction	original	0	no	0.016034	not recorded	not recorded	0.3219	constructed_law_TV, covered_value_KL, natural_KL, same_board_witness
records → prediction	original	1	no	0.019138	not recorded	not recorded	0.28963	constructed_law_TV, covered_value_KL, natural_KL, same_board_witness
records → prediction	original	2	no	0.016491	not recorded	not recorded	0.30706	constructed_law_TV, covered_value_KL, natural_KL, same_board_witness

Table S10: r11 sampling, training and registered criteria. Read directly from the hash-verified registration (r9: read-only scope). This table retains declared sampling, truncation, coverage, loss, settings and criteria, not inferred hardware. Long verbatim readings, r10’s full rational conditional-frequency table and source identities remain in the manifest-bound registration; r10 is a different task. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: `paper/data/release/prepared_r11.json.gz` (schema and SHA-256 in `release/index.json`).

registered property	value
board_counts	[22491, 1835, 109]
probe_settings/linear/C	10
probe_settings/linear/class_weight	balanced
probe_settings/linear/estimator	torch_multinomial_logistic
probe_settings/linear/fit_device	cpu
probe_settings/linear/fit_threads	1

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registered property	value
probe_settings/linear/max_iter	5,000
probe_settings/linear/solver	LBFGS_strong_wolfe
probe_settings/linear/standardization	training_mean_and_population_scale
probe_settings/linear/tol	1e-05
probe_settings/mlp64/fit_device	audit_device
probe_settings/mlp64/learning_rate	0.03
probe_settings/mlp64/updates	100
probe_settings/mlp64/width	64
settings/A_loss_normalizer	all_active_rounds
settings/AdamW_learning_rate	0.003
settings/AdamW_weight_decay	0.01
settings/CPU_threads_per_process	1
settings/audit_steps	[0, 1000, 2000, 5000, 10000, 15000, 20000]
settings/batch	256
settings/bf16_comparison_absolute_tolerances/A_vector_accuracy	0.05
settings/bf16_comparison_absolute_tolerances/B_loss_nats	0.01
settings/bf16_comparison_absolute_tolerances/natural_KL_bits	0.005
settings/device_default	cuda
settings/early_erosion_readout_steps	[1, 5, 10, 50, 100]
settings/equivalence_check/batch	256
settings/equivalence_check/nonzero_restore_update	3
settings/equivalence_check/restoration_tolerance	exact
settings/equivalence_check/rounds	8
settings/equivalence_check/updates	8
settings/equivalence_check/widths/dose	3
settings/equivalence_check/widths/erosion	1
settings/equivalence_check/widths/rarity	3
settings/equivalence_tolerances/discrete_state	exact
settings/equivalence_tolerances/optimizer_moments/atol	3e-05
settings/equivalence_tolerances/optimizer_moments/rtol	3e-05
settings/equivalence_tolerances/parameters/atol	3e-05
settings/equivalence_tolerances/parameters/rtol	3e-05
settings/evaluation_precision	fp32
settings/execution_widths/dose	3
settings/execution_widths/erosion	1
settings/execution_widths/rarity	3
settings/futility_gap_fraction	0.1
settings/futility_update	5,000
settings/maximum_CPU_processes	5
settings/precision	fp32 until the GPU 1k comparison accepts bf16
settings/rounds	8
settings/updates	20,000

Table S11: r11 complete endpoint version matrix. All recorded conditions and seeds, including equal-law controls; no endpoint is selected. Equal-law category signatures are undefined, not zero. Five significant digits for continuous values. *Full-law / long / law-row support follows Table 2. Exact-oracle errors are zero; matched initial controls and complete deciding-function calibration are in Appendix C. Census counts are rendering 0; all four renderings and saved-r9 counts are in the census table. r11 reports B_{law}.maximum_TV on all history cases. Missing measurements stay not recorded. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_B_r11.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records → prediction+sums (sums dose 100%)	equal	0	yes	4.1388e-05	not recorded	0	0.005248	B_PASS
records → prediction+sums (sums dose 100%)	equal	1	yes	0.00010222	not recorded	0	0.0055913	B_PASS
records → prediction+sums (sums dose 100%)	equal	2	yes	4.0972e-05	not recorded	0	0.0046316	B_PASS
records → prediction+sums (sums dose 100%)	original	0	no	0.0010227	298	2,944	0.2659	law_TV, rerender, swaps
records → prediction+sums (sums dose 100%)	original	1	no	0.00093253	200	8,931	0.24882	law_TV, rerender, swaps
records → prediction+sums (sums dose 100%)	original	2	no	0.0012343	406	3,619	0.27004	law_TV, rerender, swaps
records → prediction+sums (sums dose 10%)	equal	0	yes	3.9872e-05	not recorded	0	0.0052899	B_PASS
records → prediction+sums (sums dose 10%)	equal	1	yes	0.0001013	not recorded	0	0.0059298	B_PASS
records → prediction+sums (sums dose 10%)	equal	2	yes	4.3114e-05	not recorded	0	0.0058392	B_PASS
records → prediction+sums (sums dose 10%)	original	0	no	0.00063216	160	669	0.26185	law_TV, rerender, swaps
records → prediction+sums (sums dose 10%)	original	1	no	0.00083189	214	890	0.26198	law_TV, rerender, swaps
records → prediction+sums (sums dose 10%)	original	2	no	0.00068664	106	556	0.26654	law_TV, rerender, swaps
records → prediction+sums (sums dose 1%)	equal	0	yes	3.7348e-05	not recorded	0	0.0046058	B_PASS
records → prediction+sums (sums dose 1%)	equal	1	yes	0.00010275	not recorded	0	0.0065218	B_PASS
records → prediction+sums (sums dose 1%)	equal	2	yes	4.2773e-05	not recorded	0	0.0061687	B_PASS
records → prediction+sums (sums dose 1%)	original	0	no	0.00036904	81	449	0.25261	law_TV, rerender, swaps
records → prediction+sums (sums dose 1%)	original	1	no	0.0004756	82	348	0.25811	law_TV, rerender, swaps

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condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records → prediction+sums (sums dose 1%)	original	2	no	0.00061852	85	325	0.25842	law_TV, swaps
records → prediction+sums (Erosion continuation)	equal	0	yes	3.5735e-05	not recorded	0	0.0027877	B_PASS
records → prediction+sums (Erosion continuation)	equal	1	yes	9.678e-05	not recorded	0	0.0050058	B_PASS
records → prediction+sums (Erosion continuation)	equal	2	yes	3.935e-05	not recorded	0	0.0032481	B_PASS
records → prediction+sums (Erosion continuation)	original	0	no	5.0423e-05	0	0	0.25212	law_TV
records → prediction+sums (Erosion continuation)	original	1	yes	0.0001283	0	11	0.0169	B_PASS
records → prediction+sums (Erosion continuation)	original	2	no	0.00012881	0	4	0.02395	law_TV, rerender, swaps
records → prediction (Erosion continuation)	equal	0	yes	3.5714e-05	not recorded	0	0.0027865	B_PASS
records → prediction (Erosion continuation)	equal	1	yes	9.6753e-05	not recorded	0	0.0050027	B_PASS
records → prediction (Erosion continuation)	equal	2	no	3.9308e-05	not recorded	0	0.0032476	rerender
records → prediction (Erosion continuation)	original	0	no	0.00015246	32	450	0.24943	law_TV, swaps
records → prediction (Erosion continuation)	original	1	no	0.00026969	22	292	0.25779	law_TV, rerender, swaps
records → prediction (Erosion continuation)	original	2	no	0.00015369	5	31	0.23682	law_TV, rerender, swaps
records → prediction (cutoff coverage)	equal	0	yes	3.7493e-05	not recorded	0	0.0030147	B_PASS
records → prediction (cutoff coverage)	equal	1	yes	0.00012022	not recorded	0	0.0064787	B_PASS
records → prediction (cutoff coverage)	equal	2	yes	3.9549e-05	not recorded	0	0.0032866	B_PASS
records → prediction (cutoff coverage)	original	0	no	0.00014219	2	46	0.25186	law_TV, swaps
records → prediction (cutoff coverage)	original	1	no	0.00027647	25	95	0.25637	law_TV
records → prediction (cutoff coverage)	original	2	no	0.00015998	7	74	0.20843	law_TV, swaps

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condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records → prediction (decoy coverage)	equal	0	yes	3.7171e-05	not recorded	0	0.0032287	B_PASS
records → prediction (decoy coverage)	equal	1	yes	0.00012497	not recorded	0	0.0061966	B_PASS
records → prediction (decoy coverage)	equal	2	yes	4.0907e-05	not recorded	0	0.0045856	B_PASS
records → prediction (decoy coverage)	original	0	no	0.00013553	54	200	0.24966	law_TV, swaps
records → prediction (decoy coverage)	original	1	no	0.00069373	29	77	0.2571	law_TV, rerender, swaps
records → prediction (decoy coverage)	original	2	no	0.00022028	17	106	0.25388	law_TV
records → prediction (uniform sampling)	equal	0	yes	3.569e-05	not recorded	0	0.0028373	B_PASS
records → prediction (uniform sampling)	equal	1	yes	0.00010642	not recorded	0	0.0057416	B_PASS
records → prediction (uniform sampling)	equal	2	yes	3.8953e-05	not recorded	0	0.0034011	B_PASS
records → prediction (uniform sampling)	original	0	no	0.00021507	21	105	0.25012	law_TV, rerender, swaps
records → prediction (uniform sampling)	original	1	no	0.00016703	5	49	0.25216	law_TV, swaps
records → prediction (uniform sampling)	original	2	no	0.00017453	9	41	0.24988	law_TV, swaps

Table S12: r12 sampling, training and registered criteria. Read directly from the hash-verified registration (r9: read-only scope). This table retains declared sampling, truncation, coverage, loss, settings and criteria, not inferred hardware. Long verbatim readings, r10’s full rational conditional-frequency table and source identities remain in the manifest-bound registration; r10 is a different task. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/prepared_r12.json.gz (schema and SHA-256 in release/index.json).

registered property	value
settings/A_target_coefficient	1
settings/B_weights	[0.5, 0.5]
settings/CPU_threads_per_process	1
settings/MLP_fit_device	audit device
settings/audits	[0, 1000, 2000, 5000, 10000, 15000, 20000]
settings/batch	256
settings/coverage_episodes	128
settings/diagnostic_rules/TV_limit	0.02
settings/diagnostic_rules/constituent_chronology	earliest incorrect signature through first sequence violation
settings/diagnostic_rules/empty_panels	refused
settings/diagnostic_rules/recurrence	observed sequence violation, correct constituent responses through it, valid anchors
settings/diagnostic_rules/scope	descriptive, not a unique causal explanation

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registered property	value
settings/diagnostic_rules/version	2
settings/equivalence_tolerances/ discrete_state	exact
settings/equivalence_tolerances/ optimizer_moments/atol	3e-05
settings/equivalence_tolerances/ optimizer_moments/rtol	3e-05
settings/equivalence_tolerances/ parameters/atol	3e-05
settings/equivalence_tolerances/ parameters/rtol	3e-05
settings/examples_per_stratum	8
settings/execution_widths/a_supplied	1
settings/execution_widths/a_target	3
settings/execution_widths/rarity	3
settings/linear_fit_device	cpu
settings/lr	0.003
settings/maximum_processes	5
settings/natural_episodes	128
settings/precision	float32
settings/probe_settings/linear/C	10
settings/probe_settings/linear/class_ weight	balanced
settings/probe_settings/linear/ estimator	torch_multinomial_logistic
settings/probe_settings/linear/fit_ device	cpu
settings/probe_settings/linear/fit_ threads	1
settings/probe_settings/linear/max_ iter	5,000
settings/probe_settings/linear/solver	LBFGS_strong_wolfe
settings/probe_settings/linear/ standardization	training_mean_and_population_scale
settings/probe_settings/linear/tol	1e-05
settings/probe_settings/mlp64/fit_ device	audit_device
settings/probe_settings/mlp64/ learning_rate	0.03
settings/probe_settings/mlp64/ updates	100
settings/probe_settings/mlp64/width	64
settings/rounds	9
settings/steps	20,000
settings/stream_seeds	[202610110000, 202610120000, 202610130000]
settings/weight_decay	0.01

Table S13: r12 complete endpoint version matrix. All recorded conditions and seeds, including equal-law controls; no endpoint is selected. Equal-law category signatures are undefined, not zero. Five significant digits for continuous values. *Full-law / long / law-row support follows Table 2. Exact-oracle errors are zero; matched initial controls and complete deciding-function calibration are in Appendix C. Census counts are rendering 0; all four renderings and saved-r9 counts are in the census table. r11 reports B_low.maximum_TV on all history cases. Missing measurements stay not recorded. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/appendix_B_r12.json.gz (schema and SHA-256 in release/index.json).

condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records+sums → prediction	equal	0	yes	2.6087e-05	not recorded	0	0.0028994	PASS

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condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records+sums → prediction	equal	1	yes	1.1739e-05	not recorded	0	0.0019134	PASS
records+sums → prediction	equal	2	yes	0.00019948	not recorded	0	0.0071765	PASS
records+sums → prediction	original	0	yes	0.00011363	0	1	0.014557	PASS
records+sums → prediction	original	1	yes	0.00029505	1	10	0.010587	PASS
records+sums → prediction	original	2	no	0.00025691	1	51	0.012218	swaps
records → prediction+sums	equal	0	yes	2.9968e-05	not recorded	0	0.0044009	PASS
records → prediction+sums	equal	1	yes	6.9558e-05	not recorded	0	0.015836	PASS
records → prediction+sums	equal	2	yes	0.00020329	not recorded	0	0.010005	PASS
records → prediction+sums	original	0	no	0.0015898	473	1,888	0.064965	law_TV, rerender, swaps
records → prediction+sums	original	1	no	0.0020044	648	2,786	0.087776	law_TV, rerender, swaps
records → prediction+sums	original	2	no	0.0022981	815	4,272	0.044799	law_TV, rerender, swaps
records → prediction (cutoff coverage)	equal	0	yes	2.5759e-05	not recorded	0	0.0029569	PASS
records → prediction (cutoff coverage)	equal	1	yes	1.4261e-05	not recorded	0	0.0024335	PASS
records → prediction (cutoff coverage)	equal	2	yes	0.00020463	not recorded	0	0.0072873	PASS
records → prediction (cutoff coverage)	original	0	no	0.00039673	80	242	0.016568	law_TV, rerender, swaps
records → prediction (cutoff coverage)	original	1	no	0.00053515	60	395	0.025096	law_TV, rerender, swaps
records → prediction (cutoff coverage)	original	2	no	0.00062301	39	251	0.098758	law_TV, swaps
records → prediction (decoy coverage)	equal	0	yes	2.8891e-05	not recorded	0	0.0031461	PASS
records → prediction (decoy coverage)	equal	1	yes	1.3799e-05	not recorded	0	0.0022694	PASS
records → prediction (decoy coverage)	equal	2	yes	0.00020885	not recorded	0	0.0074124	PASS
records → prediction (decoy coverage)	original	0	no	0.00044239	20	183	0.028196	law_TV, rerender, swaps
records → prediction (decoy coverage)	original	1	no	0.00051612	99	510	0.018923	law_TV, rerender, swaps
records → prediction (decoy coverage)	original	2	no	0.00080244	59	472	0.021056	law_TV, rerender, swaps
records → prediction (uniform sampling)	equal	0	yes	2.6929e-05	not recorded	0	0.003395	PASS
records → prediction (uniform sampling)	equal	1	yes	1.2274e-05	not recorded	0	0.0020026	PASS

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condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records → prediction (uniform sampling)	equal	2	yes	0.00020471	not recorded	0	0.0074179	PASS
records → prediction (uniform sampling)	original	0	no	0.00052921	58	237	0.027952	law_TV, rerender, swaps
records → prediction (uniform sampling)	original	1	no	0.00054256	328	4,047	0.17174	law_TV, rerender, swaps
records → prediction (uniform sampling)	original	2	no	0.00095705	87	613	0.04589	law_TV, rerender, swaps

Table S14: r13 sampling, training and registered criteria. Read directly from the hash-verified registration (r9: read-only scope). This table retains declared sampling, truncation, coverage, loss, settings and criteria, not inferred hardware. Long verbatim readings, r10’s full rational conditional-frequency table and source identities remain in the manifest-bound registration; r10 is a different task. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/prepared_r13.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

registered property	value
settings/CPU_threads_per_process	1
settings/MLP_probe/fit_device	audit_device
settings/MLP_probe/learning_rate	0.03
settings/MLP_probe/updates	100
settings/MLP_probe/width	64
settings/audit_steps	[0, 1000, 2000, 5000, 10000, 15000, 20000]
settings/batch	256
settings/criteria/KL_bits	0.01
settings/criteria/TV	0.02
settings/criteria/witness_recovery	0.8
settings/epsilon	1e-08
settings/equivalence_tolerances/discrete_state	exact
settings/equivalence_tolerances/optimizer_moments/atol	3e-05
settings/equivalence_tolerances/optimizer_moments/rtol	3e-05
settings/equivalence_tolerances/parameters/atol	3e-05
settings/equivalence_tolerances/parameters/rtol	3e-05
settings/erosion_steps	200
settings/futility/closure	(initial_excess - current_excess) / initial_excess
settings/futility/current_excess	mean of last 100 per-batch excess_prediction_loss
settings/futility/initial_excess	B_0 - F_0
settings/futility/natural_KL_and_supervised_sums	unchanged
settings/futility/nonpositive_initial_excess	loss criterion unavailable; do not stop
settings/linear_probe/C	10
settings/linear_probe/class_weight	balanced
settings/linear_probe/estimator	torch_multinomial_logistic
settings/linear_probe/fit_device	cpu
settings/linear_probe/fit_threads	1
settings/linear_probe/max_iter	5,000
settings/linear_probe/solver	LBFGS_strong_wolfe
settings/linear_probe/standardization	training_mean_and_population_scale
settings/linear_probe/tol	1e-05

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registered property	value
settings/loss	0.5 natural-position mean + 0.5 coverage-endpoint mean; A active rounds x five slots
settings/lr	0.003
settings/maximum_processes	3
settings/numerical_encoding	(sum_twelfths/36,1-sum_twelfths/36,0,...,0)
settings/precision	float32
settings/probe_steps	[0, 5000, 20000]
settings/steps	20,000
settings/supplier_ST	hard + soft - stop_gradient(soft)
settings/wd	0.01
settings/widths/access	1
settings/widths/encoding	1
settings/widths/erosion	1
settings/widths/noise	3

Table S15: r13 complete endpoint version matrix. All recorded conditions and seeds, including equal-law controls; no endpoint is selected. Equal-law category signatures are undefined, not zero. Five significant digits for continuous values. *Full-law / long / law-row support follows Table 2. Exact-oracle errors are zero; matched initial controls and complete deciding-function calibration are in Appendix C. Census counts are rendering 0; all four renderings and saved-r9 counts are in the census table. r11 reports B_low.maximum_TV on all history cases. Missing measurements stay not recorded. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/appendix_B_r13.json.gz (schema and SHA-256 in release/index.json).

condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records →	original	0	no	0.0021068	331	2,717	0.0985	law_TV, swaps
prediction+sums	original	1	no	0.0018252	725	11,908	0.10016	law_TV, rerender, swaps
records →	original	1	no	0.00030667	0	15	0.01569	swaps
prediction+sums →	original	1	no	0.00030667	0	15	0.01569	swaps
prediction (exact supplied)	original	2	yes	0.00028061	2	24	0.010661	PASS
records+sums →	original	2	yes	0.00028061	2	24	0.010661	PASS
prediction (exact supplied)	original	2	no	0.0020459	637	2,878	0.072752	law_TV, rerender, swaps
records →	original	2	no	0.0020459	637	2,878	0.072752	law_TV, rerender, swaps
prediction+sums	original	0	no	0.00062178	259	469	0.24257	law_TV, rerender, swaps
Connected (frozen)	original	1	no	0.0005485	69	440	0.011304	rerender, swaps
Connected (frozen)	original	2	no	0.00034049	58	61	0.015699	rerender, swaps
Connected (live)	original	0	no	0.00031731	117	185	0.24683	law_TV, rerender, swaps
Connected (live)	original	1	no	0.00041094	43	180	0.25128	law_TV, rerender
Connected (live)	original	2	no	0.00029062	39	43	0.011997	rerender
records+sums →	original	0	no	0.00011384	0	38	0.013305	law_TV
prediction (exact supplied)	original	0	no	0.00011384	0	38	0.013305	law_TV
records+sums →	original	0	no	0.00083668	317	852	0.24067	law_TV, rerender, swaps
prediction (Numerical encoding)	original	0	no	0.00083668	317	852	0.24067	law_TV, rerender, swaps
records+sums →	original	1	no	0.00079477	121	1,194	0.069823	law_TV, rerender, swaps
prediction (Numerical encoding)	original	1	no	0.00079477	121	1,194	0.069823	law_TV, rerender, swaps
records+sums →	original	2	no	0.00054318	10	90	0.18902	law_TV, swaps
prediction (Numerical encoding)	original	2	no	0.00054318	10	90	0.18902	law_TV, swaps

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condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records+sums → prediction (one-hot encoding)	equal	0	yes	2.6027e-05	not recorded	0	0.0028857	CONTROL_ONLY
records+sums → prediction (one-hot encoding)	equal	1	yes	1.1735e-05	not recorded	0	0.001913	CONTROL_ONLY
records+sums → prediction (one-hot encoding)	equal	2	yes	0.00019946	not recorded	0	0.0071754	CONTROL_ONLY
records+sums → prediction (Numerical encoding)	equal	0	yes	2.6002e-05	not recorded	0	0.0028857	CONTROL_ONLY
records+sums → prediction (Numerical encoding)	equal	1	yes	1.1745e-05	not recorded	0	0.0019127	CONTROL_ONLY
records+sums → prediction (Numerical encoding)	equal	2	yes	0.00019945	not recorded	0	0.0071755	CONTROL_ONLY
AdamW	original	0	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
AdamW	original	1	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
blocked prediction gradient	original	1	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
blocked prediction gradient	original	2	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
AdamW	original	2	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
reduced-lr AdamW	original	0	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
reduced-lr AdamW	original	1	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
reduced-lr AdamW	original	2	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
SGD	original	0	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
SGD	original	1	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
SGD	original	2	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
blocked prediction gradient	original	0	not measured	not recorded	not recorded	see erosion table	not recorded	not recorded
records → prediction (sampled targets)	original	0	no	0.00052921	58	237	0.027952	law_TV, rerender, swaps
records → prediction (sampled targets)	original	1	no	0.00054256	328	4,047	0.17174	law_TV, rerender, swaps
records → prediction+sums (Probability targets)	original	1	no	0.0005186	400	2,361	0.010034	law_TV, swaps
records → prediction+sums (Probability targets)	original	2	no	0.00037991	523	2,910	0.016429	law_TV, rerender, swaps
records → prediction (sampled targets)	original	2	no	0.00095705	87	613	0.04589	law_TV, rerender, swaps
records → prediction (Probability targets)	original	0	no	7.8247e-06	2	12	0.0010311	law_TV, rerender

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condition	law	seed	B pass	KL bits	misreads	TV-error count	law TV max*	failed bars / label
records → prediction (Probability targets)	original	1	no	8.2944e-06	1	8	0.0047597	swaps
records → prediction (Probability targets)	original	2	no	0.00034879	149	324	0.1404	law_TV, rerender, swaps
records → prediction+sums (sampled targets)	original	0	no	0.0015898	473	1,888	0.064965	law_TV, rerender, swaps
records → prediction+sums (sampled targets)	original	1	no	0.0020044	648	2,786	0.087776	law_TV, rerender, swaps
records → prediction+sums (sampled targets)	original	2	no	0.0022981	815	4,272	0.044799	law_TV, rerender, swaps
records → prediction+sums (Probability targets)	original	0	no	0.00058509	473	2,897	0.037032	law_TV, rerender, swaps

Table S16: Census measures remain distinct. Counts, not percentages; original/equal laws separate. Misreads are undefined in equal-law controls. The cutoff band contains 930 boards and its complement 23,505. Complete per-sum and saved-r9 records remain hash-bound in the source tables; no rendering is substituted for another. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_B_census.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

condition	law	misread counts: range across renderings per seed	TV-error counts: range across renderings per seed
r9 records+sums → prediction (sums route only)	original	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0	s0: 1 – 1; s1: 1 – 1; s2: 34 – 34
r9 records+sums → prediction	original	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r9 records → prediction	original	s0: 8 – 8; s1: 4 – 4; s2: 3 – 3	s0: 69 – 69; s1: 25 – 25; s2: 66 – 66
r11 records → prediction+sums (sums dose 100%)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction+sums (sums dose 100%)	original	s0: 291 – 304; s1: 194 – 206; s2: 398 – 414	s0: 2,944 – 2,975; s1: 8,850 – 8,931; s2: 3,619 – 3,635
r11 records → prediction+sums (sums dose 10%)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction+sums (sums dose 10%)	original	s0: 152 – 160; s1: 214 – 240; s2: 94 – 106	s0: 638 – 670; s1: 860 – 890; s2: 523 – 586
r11 records → prediction+sums (sums dose 1%)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction+sums (sums dose 1%)	original	s0: 70 – 82; s1: 81 – 88; s2: 83 – 96	s0: 449 – 464; s1: 341 – 370; s2: 302 – 334
r11 records → prediction+sums (Erosion continuation)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction+sums (Erosion continuation)	original	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0	s0: 0 – 0; s1: 11 – 12; s2: 4 – 5
r11 records → prediction (Erosion continuation)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction (Erosion continuation)	original	s0: 29 – 37; s1: 18 – 29; s2: 4 – 8	s0: 449 – 464; s1: 278 – 297; s2: 31 – 45
r11 records → prediction (cutoff coverage)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction (cutoff coverage)	original	s0: 1 – 2; s1: 21 – 31; s2: 3 – 7	s0: 31 – 46; s1: 74 – 99; s2: 66 – 74
r11 records → prediction (decoy coverage)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction (decoy coverage)	original	s0: 40 – 54; s1: 26 – 30; s2: 14 – 18	s0: 183 – 200; s1: 77 – 89; s2: 106 – 119

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condition	law	misread counts: range across renderings per seed	TV-error counts: range across renderings per seed
r11 records → prediction (uniform sampling)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r11 records → prediction (uniform sampling)	original	s0: 17 – 22; s1: 5 – 12; s2: 7 – 13	s0: 95 – 115; s1: 47 – 64; s2: 34 – 41
r12 records+sums → prediction	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r12 records+sums → prediction	original	s0: 0 – 0; s1: 1 – 1; s2: 1 – 1	s0: 1 – 1; s1: 10 – 12; s2: 51 – 55
r12 records → prediction+sums	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r12 records → prediction+sums	original	s0: 450 – 479; s1: 639 – 678; s2: 810 – 832	s0: 1,877 – 1,919; s1: 2,786 – 2,827; s2: 4,234 – 4,272
r12 records → prediction (cutoff coverage)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r12 records → prediction (cutoff coverage)	original	s0: 80 – 98; s1: 54 – 67; s2: 33 – 39	s0: 224 – 250; s1: 376 – 399; s2: 222 – 251
r12 records → prediction (decoy coverage)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r12 records → prediction (decoy coverage)	original	s0: 19 – 23; s1: 94 – 100; s2: 59 – 83	s0: 166 – 183; s1: 508 – 511; s2: 457 – 478
r12 records → prediction (uniform sampling)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r12 records → prediction (uniform sampling)	original	s0: 58 – 60; s1: 312 – 341; s2: 84 – 90	s0: 232 – 251; s1: 4,047 – 4,138; s2: 613 – 646
r13 records → prediction+sums	original	s0: 331 – 349; s1: 725 – 740; s2: 604 – 643	s0: 2,712 – 2,740; s1: 11,895 – 11,927; s2: 2,861 – 2,889
r13 records+sums → prediction (exact supplied)	original	s0: 0 – 0; s1: 0 – 0; s2: 2 – 2	s0: 36 – 40; s1: 12 – 16; s2: 20 – 24
r13 Connected (frozen)	original	s0: 240 – 265; s1: 69 – 85; s2: 58 – 67	s0: 447 – 502; s1: 440 – 483; s2: 61 – 74
r13 Connected (live)	original	s0: 109 – 123; s1: 39 – 47; s2: 33 – 46	s0: 171 – 185; s1: 167 – 180; s2: 39 – 51
r13 records+sums → prediction (Numerical encoding)	original	s0: 309 – 332; s1: 97 – 121; s2: 6 – 10	s0: 852 – 869; s1: 1,169 – 1,200; s2: 89 – 110
r13 records+sums → prediction (one-hot encoding)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r13 records+sums → prediction (Numerical encoding)	equal	s0: not recorded; s1: not recorded; s2: not recorded	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0
r13 records → prediction (sampled targets)	original	s0: 58 – 60; s1: 312 – 341; s2: 84 – 90	s0: 232 – 251; s1: 4,047 – 4,138; s2: 613 – 646
r13 records → prediction+sums (Probability targets)	original	s0: 467 – 481; s1: 394 – 400; s2: 501 – 523	s0: 2,813 – 2,897; s1: 2,336 – 2,409; s2: 2,885 – 2,910
r13 records → prediction (Probability targets)	original	s0: 0 – 2; s1: 1 – 2; s2: 149 – 176	s0: 9 – 16; s1: 8 – 14; s2: 324 – 352
r13 records → prediction+sums (sampled targets)	original	s0: 450 – 479; s1: 639 – 678; s2: 810 – 832	s0: 1,877 – 1,919; s1: 2,786 – 2,827; s2: 4,234 – 4,272

Table S17: Exact support: sums, supplier, local histories and saved-r9 cases. Slot components and whole-vector counts are separate. Denominators follow the named quantity; a frozen supplier is not the same as an approximate sums output. Saved-r9 rows are constructed historical failing renderings, not new random samples. Missing local-history audits are not zero errors. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/appendix_B_support.json.gz (schema and SHA-256 in release/index.json).

run / seed / law	exact vector / local audited quantities
r11 records → prediction+sums (sums dose 100%)/0 equal	A_components / vector / correct: 18,963; A_components / vector / total: 24,435
r11 records → prediction+sums (sums dose 100%)/1 equal	A_components / vector / correct: 19,562; A_components / vector / total: 24,435
r11 records → prediction+sums (sums dose 100%)/2 equal	A_components / vector / correct: 20,295; A_components / vector / total: 24,435
r11 records → prediction+sums (sums dose 100%)/0 original	A_components / vector / correct: 20,012; A_components / vector / total: 24,435

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run / seed / law	exact vector / local audited quantities
r13 records → prediction (sampled targets)/1 original	sums / vector / correct: 0; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction+sums (Probability targets)/1 original	sums / vector / correct: 19,527; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction+sums (Probability targets)/2 original	sums / vector / correct: 19,513; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction (sampled targets)/2 original	sums / vector / correct: 0; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction (Probability targets)/0 original	sums / vector / correct: 0; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction (Probability targets)/1 original	sums / vector / correct: 0; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction (Probability targets)/2 original	sums / vector / correct: 0; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction+sums (sampled targets)/0 original	sums / vector / correct: 19,269; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction+sums (sampled targets)/1 original	sums / vector / correct: 20,516; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction+sums (sampled targets)/2 original	sums / vector / correct: 19,544; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435
r13 records → prediction+sums (Probability targets)/0 original	sums / vector / correct: 19,921; sums / vector / total: 24,435; supplier / vector / correct: 0; supplier / vector / total: 24,435

Table S18: Continuous law errors, by case. L/H and N1–N2 form the short panel; N3–N8 form the long panel. The r12 long panel is trained, not unseen-length extrapolation. The original-task TV bar is 0.02; r10 original-law law rows use 0.05 and equal-law rows 0.02. The exact ceiling is zero. r10 law rows are a different all-slots task, not the same denominator as r8/r11–r13 histories. Mean TV is not reconstructed when absent from the saved summary. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/appendix_B_laws.json.gz (schema and SHA-256 in <release/index.json>).

condition / law	panel	presentations per seed	case-mean TV ranges (all seeds)	worst TV seeds 0/1/2	long worst TV seeds 0/1/2
r10 records+sums → prediction (sums route only) equal	eight_hold	90	not recorded	0.0041717/ 0.0019756/ 0.0032849	not recorded
r10 records+sums → prediction (sums route only) equal	heldout_board	12,000	not recorded	0.0041739/ 0.0019765/ 0.0032851	not recorded
r10 records+sums → prediction (sums route only) equal	heldout_rendering	60,000	not recorded	0.004174/ 0.001977/ 0.0032851	not recorded
r10 records+sums → prediction (sums route only) equal	law_rows	180	not recorded	0.0041717/ 0.0019756/ 0.0032849	not recorded
r10 records+sums → prediction (sums route only) equal	natural	60,000	not recorded	0.004174/ 0.001977/ 0.0032851	not recorded
r10 records+sums → prediction (sums route only) equal	witness	20	not recorded	0.0041716/ 0.0019756/ 0.0032849	not recorded
r10 records+sums → prediction (sums route only) original	eight_hold	90	not recorded	0.086635/ 0.12994/ 0.094673	not recorded
r10 records+sums → prediction (sums route only) original	heldout_board	12,000	not recorded	0.20385/ 0.15108/0.16369	not recorded
r10 records+sums → prediction (sums route only) original	heldout_rendering	60,000	not recorded	0.23835/ 0.25233/0.24308	not recorded
r10 records+sums → prediction (sums route only) original	law_rows	180	not recorded	0.12242/ 0.079414/ 0.10517	not recorded

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condition / law	panel	presentations per seed	case-mean TV ranges (all seeds)	worst TV seeds 0/1/2	long worst TV seeds 0/1/2
r10 records+sums → prediction (sums route only) original	natural	60,000	not recorded	0.23835/ 0.25233/0.24308	not recorded
r10 records+sums → prediction (sums route only) original	witness	20	not recorded	0.074905/ 0.083315/ 0.094673	not recorded
r10 records → prediction+sums equal	eight_hold	90	not recorded	0.004654/ 0.0017789/ 0.0034453	not recorded
r10 records → prediction+sums equal	heldout_board	12,000	not recorded	0.0061441/ 0.0054889/ 0.0046655	not recorded
r10 records → prediction+sums equal	heldout_ rendering	60,000	not recorded	0.0063493/ 0.0061142/ 0.0054515	not recorded
r10 records → prediction+sums equal	law_rows	180	not recorded	0.005519/ 0.0020335/ 0.0040652	not recorded
r10 records → prediction+sums equal	natural	60,000	not recorded	0.0063719/ 0.0067935/ 0.0055263	not recorded
r10 records → prediction+sums equal	witness	20	not recorded	0.004654/ 0.001763/ 0.0033482	not recorded
r10 records → prediction+sums original	eight_hold	90	not recorded	0.21721/ 0.12493/0.17067	not recorded
r10 records → prediction+sums original	heldout_board	12,000	not recorded	0.37209/ 0.29582/0.36441	not recorded
r10 records → prediction+sums original	heldout_ rendering	60,000	not recorded	0.41321/ 0.44997/0.48577	not recorded
r10 records → prediction+sums original	law_rows	180	not recorded	0.27737/ 0.25519/0.23295	not recorded
r10 records → prediction+sums original	natural	60,000	not recorded	0.42721/ 0.43376/0.48519	not recorded
r10 records → prediction+sums original	witness	20	not recorded	0.20383/ 0.11576/0.16125	not recorded
r10 records → prediction equal	eight_hold	90	not recorded	0.0061219/ 0.0023327/ 0.0032967	not recorded
r10 records → prediction equal	heldout_board	12,000	not recorded	0.0097939/ 0.13205/ 0.0032921	not recorded
r10 records → prediction equal	heldout_ rendering	60,000	not recorded	0.028351/ 0.14421/ 0.0032966	not recorded
r10 records → prediction equal	law_rows	180	not recorded	0.0046541/ 0.042784/ 0.0032866	not recorded
r10 records → prediction equal	natural	60,000	not recorded	0.016887/ 0.14656/ 0.0033036	not recorded
r10 records → prediction equal	witness	20	not recorded	0.004758/ 0.0020253/ 0.0032877	not recorded
r10 records → prediction original	eight_hold	90	not recorded	0.14088/ 0.15646/0.17271	not recorded
r10 records → prediction original	heldout_board	12,000	not recorded	0.30396/ 0.30772/0.30336	not recorded
r10 records → prediction original	heldout_ rendering	60,000	not recorded	0.36339/0.3819/ 0.43667	not recorded
r10 records → prediction original	law_rows	180	not recorded	0.3219/0.28963/ 0.30706	not recorded
r10 records → prediction original	natural	60,000	not recorded	0.33005/0.3388/ 0.40158	not recorded

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condition / law	panel	presentations per seed	case-mean TV ranges (all seeds)	worst TV seeds 0/1/2	long worst TV seeds 0/1/2
r10 records → prediction original	witness	20	not recorded	0.13041/ 0.15646/0.17271	not recorded
r11 records → prediction+sums (sums dose 100%) equal	short / long	192 / 384	short 0.0026749 – 0.0049599; long 0.0029476 – 0.00509	0.0035675/ 0.0055413/ 0.0045952	0.005248/ 0.0055913/ 0.0046316
r11 records → prediction+sums (sums dose 100%) original	short / long	192 / 384	short 0.0038291 – 0.022196; long 0.012391 – 0.058066	0.23211/ 0.047024/ 0.18522	0.2659/0.24882/ 0.27004
r11 records → prediction+sums (sums dose 10%) equal	short / long	192 / 384	short 0.0027822 – 0.0049704; long 0.0029866 – 0.0051381	0.0046949/ 0.0055218/ 0.0054278	0.0052899/ 0.0059298/ 0.0058392
r11 records → prediction+sums (sums dose 10%) original	short / long	192 / 384	short 0.0065999 – 0.013873; long 0.0093113 – 0.042768	0.076521/ 0.21589/0.19524	0.26185/ 0.26198/0.26654
r11 records → prediction+sums (sums dose 1%) equal	short / long	192 / 384	short 0.0026845 – 0.0049643; long 0.0028501 – 0.0051419	0.0036955/ 0.0058059/ 0.005799	0.0046058/ 0.0065218/ 0.0061687
r11 records → prediction+sums (sums dose 1%) original	short / long	192 / 384	short 0.0052516 – 0.010872; long 0.008599 – 0.049405	0.16126/ 0.076986/ 0.11075	0.25261/ 0.25811/0.25842
r11 records → prediction+sums (Erosion continuation) equal	short / long	192 / 384	short 0.0027869 – 0.0050022; long 0.0027854 – 0.0050046	0.0027877/ 0.0050028/ 0.0032481	0.0027875/ 0.0050058/ 0.003248
r11 records → prediction+sums (Erosion continuation) original	short / long	192 / 384	short 0.0011696 – 0.007621; long 0.0040712 – 0.019293	0.010885/ 0.016244/ 0.02395	0.25212/0.0169/ 0.023262
r11 records → prediction (Erosion continuation) equal	short / long	192 / 384	short 0.0027863 – 0.0050021; long 0.0027863 – 0.0050026	0.0027864/ 0.0050022/ 0.0032476	0.0027865/ 0.0050027/ 0.0032476
r11 records → prediction (Erosion continuation) original	short / long	192 / 384	short 0.00089769 – 0.0087451; long 0.0047204 – 0.010597	0.010811/ 0.055624/ 0.013042	0.24943/ 0.25779/0.23682
r11 records → prediction (cutoff coverage) equal	short / long	192 / 384	short 0.0027902 – 0.0048927; long 0.0028481 – 0.0059251	0.0029098/ 0.0051907/ 0.003269	0.0030147/ 0.0064787/ 0.0032866
r11 records → prediction (cutoff coverage) original	short / long	192 / 384	short 0.0023639 – 0.0067305; long 0.0049594 – 0.023397	0.040826/ 0.013095/ 0.020684	0.25186/ 0.25637/0.20843
r11 records → prediction (decoy coverage) equal	short / long	192 / 384	short 0.0028003 – 0.0050136; long 0.0028627 – 0.0060584	0.0031166/ 0.0051079/ 0.0040236	0.0032287/ 0.0061966/ 0.0045856
r11 records → prediction (decoy coverage) original	short / long	192 / 384	short 0.0031058 – 0.021927; long 0.0047785 – 0.047953	0.0099633/ 0.2571/0.011653	0.24966/ 0.25686/0.25388
r11 records → prediction (uniform sampling) equal	short / long	192 / 384	short 0.0027887 – 0.0050242; long 0.0027975 – 0.0055232	0.0028107/ 0.0051969/ 0.0033609	0.0028373/ 0.0057416/ 0.0034011
r11 records → prediction (uniform sampling) original	short / long	192 / 384	short 0.0028587 – 0.0066598; long 0.0048924 – 0.028613	0.05598/ 0.011085/ 0.02274	0.25012/ 0.25216/0.24988
r12 records+sums → prediction equal	short / long	192 / 384	short 0.0019128 – 0.0071754; long 0.0019128 – 0.007176	0.0028898/ 0.0019132/ 0.007176	0.0028994/ 0.0019134/ 0.0071765
r12 records+sums → prediction original	short / long	192 / 384	short 0.0019765 – 0.010642; long 0.0038941 – 0.0097004	0.015676/ 0.0093861/ 0.01258	0.014557/ 0.010587/ 0.012218
r12 records → prediction+sums equal	short / long	192 / 384	short 0.0019278 – 0.0070919; long 0.0024947 – 0.0076936	0.004377/ 0.0098577/ 0.010771	0.0044009/ 0.015836/ 0.010005
r12 records → prediction+sums original	short / long	192 / 384	short 0.008027 – 0.022173; long 0.0079901 – 0.018263	0.049233/ 0.13295/0.03955	0.064965/ 0.087776/ 0.044799

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condition / law	panel	presentations per seed	case-mean TV ranges (all seeds)	worst TV seeds 0/1/2	long worst TV seeds 0/1/2
r12 records → prediction (cutoff coverage) equal	short / long	192 / 384	short 0.0018449 – 0.0070508; long 0.0019628 – 0.0070866	0.0029008/ 0.0021976/ 0.0071481	0.0029569/ 0.0024335/ 0.0072873
r12 records → prediction (cutoff coverage) original	short / long	192 / 384	short 0.004765 – 0.012261; long 0.0048786 – 0.011082	0.045729/ 0.051614/ 0.051641	0.016568/ 0.025096/ 0.098758
r12 records → prediction (decoy coverage) equal	short / long	192 / 384	short 0.0020236 – 0.0071865; long 0.0020288 – 0.0073008	0.0031639/ 0.0022769/ 0.0072713	0.0031461/ 0.0022694/ 0.0074124
r12 records → prediction (decoy coverage) original	short / long	192 / 384	short 0.0051834 – 0.011809; long 0.0047664 – 0.01477	0.034469/ 0.035257/ 0.046889	0.028196/ 0.018923/ 0.021056
r12 records → prediction (uniform sampling) equal	short / long	192 / 384	short 0.0019063 – 0.0071816; long 0.0019321 – 0.0073279	0.003195/ 0.0019451/ 0.0072272	0.003395/ 0.0020026/ 0.0074179
r12 records → prediction (uniform sampling) original	short / long	192 / 384	short 0.0052395 – 0.013936; long 0.0048822 – 0.013377	0.033385/ 0.020324/ 0.066427	0.027952/ 0.17174/0.04589
r13 records → prediction+sums original	short / long	192 / 384	short 0.0069933 – 0.025547; long 0.0071803 – 0.01413	0.081309/ 0.044418/ 0.047182	0.0985/0.10016/ 0.072752
r13 records+sums → prediction (exact supplied) original	short / long	192 / 384	short 0.0026929 – 0.010788; long 0.0041718 – 0.0097128	0.022749/ 0.012822/ 0.011454	0.013305/ 0.01569/ 0.010661
r13 Connected (frozen) original	short / long	192 / 384	short 0.0012879 – 0.010469; long 0.0056155 – 0.0098562	0.013796/ 0.014747/ 0.013029	0.24257/ 0.011304/ 0.015699
r13 Connected (live) original	short / long	192 / 384	short 0.0022243 – 0.010608; long 0.0053861 – 0.012634	0.025362/ 0.015556/ 0.01333	0.24683/ 0.25128/ 0.011997
r13 records+sums → prediction (Numerical encoding) original	short / long	192 / 384	short 0.0025803 – 0.018088; long 0.0044912 – 0.012229	0.028812/ 0.042917/ 0.021251	0.24067/ 0.069823/ 0.18902
r13 records+sums → prediction (one-hot encoding) equal	short / long	192 / 384	short 0.0019128 – 0.0071753; long 0.0019123 – 0.0071753	0.0028857/ 0.0019131/ 0.0071755	0.0028857/ 0.001913/ 0.0071754
r13 records+sums → prediction (Numerical encoding) equal	short / long	192 / 384	short 0.0019127 – 0.0071755; long 0.0019126 – 0.0071755	0.0028857/ 0.0019127/ 0.0071755	0.0028857/ 0.0019127/ 0.0071755
r13 records → prediction (sampled targets) original	short / long	192 / 384	short 0.0052395 – 0.013936; long 0.0048822 – 0.013377	0.033385/ 0.020324/ 0.066427	0.027952/ 0.17174/0.04589
r13 records → prediction+sums (Probability targets) original	short / long	192 / 384	short 0.001634 – 0.0091321; long 0.0019609 – 0.0029218	0.02855/ 0.035266/ 0.023742	0.037032/ 0.010034/ 0.016429
r13 records → prediction (Probability targets) original	short / long	192 / 384	short 0.00020485 – 0.0014381; long 0.000297 – 0.003707	0.022169/ 0.013685/ 0.0045307	0.0010311/ 0.0047597/ 0.1404
r13 records → prediction+sums (sampled targets) original	short / long	192 / 384	short 0.008027 – 0.022173; long 0.0079901 – 0.018263	0.049233/ 0.13295/0.03955	0.064965/ 0.087776/ 0.044799

Table S19: Current versus remembered value exposure. Actual online-training exposures, not independent sample sizes or inference counts. The current and remembered value arrays are separate. Readability support is board-identity held out; values missing from the registered fit/score panel, especially 28–36, are supplementary rendering support, not established by the registered probe gain. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_B_r10_coverage.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

run / seed / law	current occurrences s1–s5	remembered occurrences s1–s5	current values covered	remembered values covered
records+sums → prediction (sums route only)/0 equal	8,191,214/8,196,282/ 8,190,250/8,190,488/ 8,191,766	8,191,214/8,196,282/ 8,190,250/8,190,488/ 8,191,766	36/36/36/36/ 36	36/36/36/36/ 36
records+sums → prediction (sums route only)/1 equal	8,189,041/8,190,212/ 8,199,123/8,189,913/ 8,191,711	8,189,041/8,190,212/ 8,199,123/8,189,913/ 8,191,711	36/36/36/36/ 36	36/36/36/36/ 36
records+sums → prediction (sums route only)/2 equal	8,190,289/8,193,352/ 8,188,704/8,192,479/ 8,195,176	8,190,289/8,193,352/ 8,188,704/8,192,479/ 8,195,176	36/36/36/36/ 36	36/36/36/36/ 36
records+sums → prediction (sums route only)/0 original	8,191,238/8,191,265/ 8,189,168/8,196,137/ 8,192,192	8,191,238/8,191,265/ 8,189,168/8,196,137/ 8,192,192	36/36/36/36/ 36	36/36/36/36/ 36
records+sums → prediction (sums route only)/1 original	8,192,442/8,190,242/ 8,191,352/8,190,986/ 8,194,978	8,192,442/8,190,242/ 8,191,352/8,190,986/ 8,194,978	36/36/36/36/ 36	36/36/36/36/ 36
records+sums → prediction (sums route only)/2 original	8,187,664/8,193,707/ 8,191,278/8,192,114/ 8,195,237	8,187,664/8,193,707/ 8,191,278/8,192,114/ 8,195,237	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction+sums/0 equal	8,191,214/8,196,282/ 8,190,250/8,190,488/ 8,191,766	8,191,214/8,196,282/ 8,190,250/8,190,488/ 8,191,766	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction+sums/1 equal	8,189,041/8,190,212/ 8,199,123/8,189,913/ 8,191,711	8,189,041/8,190,212/ 8,199,123/8,189,913/ 8,191,711	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction+sums/2 equal	8,190,289/8,193,352/ 8,188,704/8,192,479/ 8,195,176	8,190,289/8,193,352/ 8,188,704/8,192,479/ 8,195,176	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction+sums/0 original	8,191,238/8,191,265/ 8,189,168/8,196,137/ 8,192,192	8,191,238/8,191,265/ 8,189,168/8,196,137/ 8,192,192	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction+sums/1 original	8,192,442/8,190,242/ 8,191,352/8,190,986/ 8,194,978	8,192,442/8,190,242/ 8,191,352/8,190,986/ 8,194,978	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction+sums/2 original	8,187,664/8,193,707/ 8,191,278/8,192,114/ 8,195,237	8,187,664/8,193,707/ 8,191,278/8,192,114/ 8,195,237	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction/0 equal	8,191,214/8,196,282/ 8,190,250/8,190,488/ 8,191,766	8,191,214/8,196,282/ 8,190,250/8,190,488/ 8,191,766	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction/1 equal	8,189,041/8,190,212/ 8,199,123/8,189,913/ 8,191,711	8,189,041/8,190,212/ 8,199,123/8,189,913/ 8,191,711	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction/2 equal	8,190,289/8,193,352/ 8,188,704/8,192,479/ 8,195,176	8,190,289/8,193,352/ 8,188,704/8,192,479/ 8,195,176	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction/0 original	8,191,238/8,191,265/ 8,189,168/8,196,137/ 8,192,192	8,191,238/8,191,265/ 8,189,168/8,196,137/ 8,192,192	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction/1 original	8,192,442/8,190,242/ 8,191,352/8,190,986/ 8,194,978	8,192,442/8,190,242/ 8,191,352/8,190,986/ 8,194,978	36/36/36/36/ 36	36/36/36/36/ 36
records → prediction/2 original	8,187,664/8,193,707/ 8,191,278/8,192,114/ 8,195,237	8,187,664/8,193,707/ 8,191,278/8,192,114/ 8,195,237	36/36/36/36/ 36	36/36/36/36/ 36

Table S20: Deciding-function calibration: positives, negatives and support. Named scalar calibration summaries, with exact counts and original/equal law-aware decisions. A successful calibration is not a successful trained model. Unsupported/empty nulls do not pass; the r10 trained positive is a fixed-endpoint condition. Floors and exact-oracle ceilings are retained in the named condition rows. Per-presentation and per-slot arrays remain in the hash-verified full calibration records; slot floors are in the probe table. Five significant digits for continuous values. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_C_calibration.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

study	law	positive / null	deciding quantity	pass	support	law max / rates
r8	original	current_board_null	B deciding criteria	no	576	0.1875
r8	original	forced_A	B deciding criteria	yes	576	0
r8	original	forced_raw	B deciding criteria	yes	576	0
r8	equal	current_board_null	B deciding criteria	yes	576	2.9802e-08
r8	equal	forced_A	B deciding criteria	yes	576	2.9802e-08
r8	equal	forced_raw	B deciding criteria	yes	576	2.9802e-08
r8	both	oracle	raw linear sum calibration (≥ 0.99 ; convergence)	yes	512	512/512/512/511/511 / 512
r8	both	update0	raw linear sum calibration (≥ 0.99 ; convergence)	no	512	187/167/121/153/152 / 512
r10	original	exact_oracle	B deciding criteria	yes	180 law rows	4.4703e-08
r10	original	mode_blind	B deciding criteria	no	180 law rows	0.48635
r10	original	untrained	B deciding criteria	no	180 law rows	0.42896
r10	equal	exact_oracle	B deciding criteria	yes	180 law rows	1.4901e-08
r10	equal	mode_blind	B deciding criteria	yes	180 law rows	0
r10	equal	untrained	B deciding criteria	no	180 law rows	0.065678
r10	original	slot 1: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.1123
r10	original	slot 2: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.053711
r10	original	slot 3: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0
r10	original	slot 4: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.18164
r10	original	slot 5: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.18164
r10	original	planted contamination	other-slot changes detected	yes	1,024	949/950/926/968/959
r10	both	oracle	raw linear sum calibration (≥ 0.99 ; convergence)	yes	512	512/512/511/512/511 / 512
r10	both	untrained	raw linear sum calibration (≥ 0.99 ; convergence)	no	512	173/154/127/177/147 / 512
r11	original	current_board_null	B deciding criteria	no	576	0.1875
r11	original	oracle_A	B deciding criteria	yes	576	0
r11	original	oracle_raw	B deciding criteria	yes	576	0
r11	equal	current_board_null	B deciding criteria	yes	576	2.9802e-08
r11	equal	oracle_A	B deciding criteria	yes	576	2.9802e-08
r11	equal	oracle_raw	B deciding criteria	yes	576	2.9802e-08
r11	both	oracle	raw linear sum calibration (≥ 0.99 ; convergence)	yes	512	512/512/512/511/511 / 512
r11	both	untrained	raw linear sum calibration (≥ 0.99 ; convergence)	no	512	186/166/120/154/152 / 512
r12	original	current_board_null	B deciding criteria	no	576	0.1875
r12	original	forced_A_oracle	B deciding criteria	yes	576	0
r12	original	oracle	B deciding criteria	yes	576	0
r12	equal	current_board_null	B deciding criteria	yes	576	2.9802e-08
r12	equal	forced_A_oracle	B deciding criteria	yes	576	2.9802e-08
r12	equal	oracle	B deciding criteria	yes	576	2.9802e-08

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study	law	positive / null	deciding quantity	pass	support	law max / rates
r12	both	oracle	raw linear sum calibration (≥ 0.99 ; convergence)	yes	512	512/512/512/511/511 / 512
r12	both	untrained	raw linear sum calibration (≥ 0.99 ; convergence)	no	512	186/166/120/154/152 / 512
r13	original	board_only	B deciding criteria	no	576 history endpoints	0.1875
r13	original	oracle	B deciding criteria	yes	576 history endpoints	0
r13	equal	board_only	B deciding criteria	yes	576 history endpoints	2.9802e-08
r13	equal	oracle	B deciding criteria	yes	576 history endpoints	2.9802e-08

Table S21: Interchange support, calibration and selectivity are separate. Fixed prediction-independent pair lists, 1,024 per slot; no resampling. Conditional donor-follow uses accurate unpatched pairs as its denominator, not all eligible pairs. Positive validity, supported negative calibration and zero-spillover selectivity are separate gates. The oracle ceiling is exact donor-follow with zero other-slot changes; no-op/shuffled floors are in Table C calibration. Equal-law rows are control-only. a_forced’s shuffled raw negative is not applicable. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/appendix_C_interchange.json.gz (schema and SHA-256 in release/index.json).

run / seed / law	accurate pairs s1–s5	donor-follow counts	other-slot changes	positive validity	use decision
records+sums → prediction (sums route only)/0 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	0/0/0/0/0	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records+sums → prediction (sums route only)/1 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	0/0/0/0/0	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records+sums → prediction (sums route only)/2 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	0/0/0/0/0	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records+sums → prediction (sums route only)/0 original	875/844/710/ 841/947	866/842/693/ 835/941	0/0/0/0/0	VALID	KNOWN_USE_ POSITIVE_ VALID
records+sums → prediction (sums route only)/1 original	987/989/974/ 966/881	975/986/969/ 964/878	0/0/0/0/0	VALID	KNOWN_USE_ POSITIVE_ VALID
records+sums → prediction (sums route only)/2 original	857/883/926/ 940/924	851/877/923/ 936/921	0/0/0/0/0	VALID	KNOWN_USE_ POSITIVE_ VALID
records → prediction+sums/0 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	181/293/333/ 269/181	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records → prediction+sums/1 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	257/320/263/ 504/334	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records → prediction+sums/2 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	176/186/197/ 175/219	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records → prediction+sums/0 original	242/201/233/ 124/224	218/146/160/79/ 198	346/261/279/ 235/279	VALID	CALIBRATED; UNCALIBRATED
records → prediction+sums/1 original	207/326/141/90/ 161	163/286/121/72/ 125	280/193/188/ 284/188	VALID	CALIBRATED; UNCALIBRATED
records → prediction+sums/2 original	118/198/170/ 121/164	101/146/154/ 105/110	243/220/294/ 248/286	VALID	CALIBRATED; UNCALIBRATED
records → prediction/0 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	1,250/1,632/ 1,699/1,778/ 1,872	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY

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run / seed / law	accurate pairs s1-s5	donor-follow counts	other-slot changes	positive validity	use decision
records → prediction/1 equal	1,020/1,024/ 1,017/1,024/ 1,024	0/0/0/0/0	2,461/463/ 1,420/905/1,562	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records → prediction/2 equal	1,024/1,024/ 1,024/1,024/ 1,024	0/0/0/0/0	1,632/1,525/ 1,936/2,306/ 1,993	NOT_ APPLICABLE	EQUAL_LAW_ CONTROL_ ONLY
records → prediction/0 original	702/433/374/ 705/409	601/375/297/ 511/338	341/223/275/ 436/276	VALID	CALIBRATED
records → prediction/1 original	464/465/445/ 689/517	381/407/413/ 613/464	361/206/218/ 354/200	VALID	CALIBRATED; UNCALIBRATED
records → prediction/2 original	561/614/465/ 587/675	518/527/298/ 523/651	353/283/225/ 329/325	VALID	CALIBRATED

Table S22: Full probes: carrier, reader, controls and paired intervals. Every slot and seed for both carriers and readout families in r8/r10; 512 held-out board identities per cell. r8 historical heldout fields retain their saved five-decimal precision; raw-only outline endpoints use the ledger’s unrounded fields. Gain intervals are paired board-bootstrap intervals, not endpoint intervals or seed variation. Rare r10 values remain outside registered readability support; see the per-value coverage table. MLP fits have a fixed budget, not a solver-convergence assessment. Analytical reduction only; exact per-cell and per-update rows remain released. Printed sum-reader ranges retain seed identities; category readers remain in the detailed file. Gain CI envelopes span the five saved slot intervals and are not new confidence intervals. Floor ranges span seeds and slots. Detailed records: paper/data/release/appendix_E_probes.json.gz (schema and SHA-256 in release/index.json).

condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
r8 records → prediction equal	raw linear	s0: 0.23633 – 0.36523; s1: 0.25781 – 0.36719; s2: 0.25586 – 0.35938	s0: 0.42578 – 0.61133; s1: 0.41016 – 0.5957; s2: 0.27539 – 0.59961	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 0.99805 – 1	s0: 0.0078125 – 0.42773; s1: 0.097656 – 0.39263; s2: -0.033203 – 0.38867	all seeds
r8 records → prediction equal	raw mlp64	s0: 0.57812 – 0.62695; s1: 0.56445 – 0.68555; s2: 0.55664 – 0.6543	s0: 0.37305 – 0.58398; s1: 0.4707 – 0.61914; s2: 0.47266 – 0.625	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 1 – 1	s0: -0.26177 – 0.023486; s1: -0.17583 – 0.033203; s2: -0.18555 – 0.046875	fixed budget
r8 records → prediction equal	upper linear	s0: 0.27539 – 0.38086; s1: 0.2832 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.30273 – 0.50977; s1: 0.39844 – 0.57422; s2: 0.2207 – 0.58789	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 0.99805 – 1	s0: -0.12891 – 0.26562; s1: 0.042969 – 0.34375; s2: -0.095703 – 0.36133	all seeds
r8 records → prediction equal	upper mlp64	s0: 0.53906 – 0.55664; s1: 0.52539 – 0.60547; s2: 0.49805 – 0.61523	s0: 0.33984 – 0.41602; s1: 0.33984 – 0.40625; s2: 0.33984 – 0.40625	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 0.99805 – 1	s0: -0.25 – -0.089844; s1: -0.26758 – -0.095703; s2: -0.29883 – -0.10352	fixed budget
r8 records → prediction original	raw linear	s0: 0.23633 – 0.36523; s1: 0.25781 – 0.36719; s2: 0.25586 – 0.35938	s0: 0.06445 – 0.96875; s1: 0.06641 – 0.9707; s2: 0.05078 – 0.97656	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 0.99805 – 1	s0: -0.27935 – 0.64648; s1: -0.33594 – 0.70508; s2: -0.35161 – 0.75591	all seeds

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
r8 records → prediction original	raw mlp64	s0: 0.57812 – 0.62695; s1: 0.56445 – 0.68555; s2: 0.55664 – 0.6543	s0: 0.35156 – 0.98047; s1: 0.35352 – 0.97852; s2: 0.35156 – 0.97656	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 1 – 1	s0: -0.30078 – 0.43164; s1: -0.33203 – 0.41211; s2: -0.33008 – 0.46099	fixed budget
r8 records → prediction original	upper linear	s0: 0.27539 – 0.38086; s1: 0.2832 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.04102 – 0.9668; s1: 0.04492 – 0.97656; s2: 0.04492 – 0.97656	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 0.99805 – 1	s0: -0.32617 – 0.62695; s1: -0.38672 – 0.67192; s2: -0.36328 – 0.74414	all seeds
r8 records → prediction original	upper mlp64	s0: 0.53906 – 0.55664; s1: 0.52539 – 0.60547; s2: 0.49805 – 0.61523	s0: 0.35547 – 0.5918; s1: 0.35547 – 0.67188; s2: 0.35547 – 0.70898	majority 0.33984 – 0.40625; shuffled 0.16992 – 0.20703; oracle 0.99805 – 1	s0: -0.22266 – 0.085938; s1: -0.23828 – 0.15044; s2: -0.29883 – 0.25781	fixed budget
r10 records+sums → prediction (sums route only) equal	raw linear sum	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	majority 0.36328 – 0.38672; shuffled 0.0039062 – 0.021484; oracle 0.99805 – 1	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0	all seeds
r10 records+sums → prediction (sums route only) equal	raw mlp64 sum	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	majority 0.36328 – 0.38672; shuffled 0.27734 – 0.33594; oracle 1 – 1	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0	fixed budget
r10 records+sums → prediction (sums route only) equal	upper linear sum	s0: 0.63086 – 0.73828; s1: 0.64258 – 0.72656; s2: 0.67383 – 0.76562	s0: 0.042969 – 0.066406; s1: 0.025391 – 0.056641; s2: 0.037109 – 0.070312	majority 0.36328 – 0.38672; shuffled 0.0078125 – 0.025391; oracle 0.99805 – 1	s0: -0.71484 – -0.54297; s1: -0.74219 – -0.53711; s2: -0.73633 – -0.57222	all seeds
r10 records+sums → prediction (sums route only) equal	upper mlp64 sum	s0: 0.66797 – 0.75781; s1: 0.66797 – 0.73242; s2: 0.68945 – 0.77344	s0: 0.28906 – 0.30859; s1: 0.26562 – 0.32031; s2: 0.27148 – 0.30078	majority 0.36328 – 0.38672; shuffled 0.21875 – 0.31836; oracle 1 – 1	s0: -0.502 – -0.32026; s1: -0.50586 – -0.33203; s2: -0.55078 – -0.3457	fixed budget
r10 records+sums → prediction (sums route only) original	raw linear sum	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	majority 0.36328 – 0.38672; shuffled 0.0039062 – 0.021484; oracle 0.99805 – 1	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0	all seeds
r10 records+sums → prediction (sums route only) original	raw mlp64 sum	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	majority 0.36328 – 0.38672; shuffled 0.27734 – 0.33594; oracle 1 – 1	s0: 0 – 0; s1: 0 – 0; s2: 0 – 0	fixed budget
r10 records+sums → prediction (sums route only) original	upper linear sum	s0: 0.63086 – 0.73828; s1: 0.64258 – 0.72656; s2: 0.67383 – 0.76562	s0: 0.51562 – 0.68555; s1: 0.53711 – 0.68945; s2: 0.54688 – 0.76953	majority 0.36328 – 0.38672; shuffled 0.0058594 – 0.027344; oracle 0.99805 – 1	s0: -0.27344 – 0.066406; s1: -0.23438 – 0.097656; s2: -0.25005 – 0.054736	all seeds

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
r10 records+sums → prediction (sums route only) original	upper mlp64 sum	s0: 0.66797 – 0.75781; s1: 0.66797 – 0.73242; s2: 0.68945 – 0.77344	s0: 0.53906 – 0.69531; s1: 0.56445 – 0.75; s2: 0.67578 – 0.79492	majority 0.36328 – 0.38672; shuffled 0.16602 – 0.23438; oracle 1 – 1	s0: -0.25781 – 0.0625; s1: -0.21289 – 0.13096; s2: -0.13672 – 0.11133	fixed budget
r10 records → prediction+sums equal	raw linear sum	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	s0: 0.91602 – 0.95703; s1: 0.93164 – 0.96094; s2: 0.93164 – 0.95117	majority 0.36328 – 0.38672; shuffled 0.0078125 – 0.021484; oracle 0.99805 – 1	s0: 0.54883 – 0.74609; s1: 0.55078 – 0.74023; s2: 0.52344 – 0.78716	all seeds
r10 records → prediction+sums equal	raw mlp64 sum	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	s0: 0.91211 – 0.96094; s1: 0.92969 – 0.9707; s2: 0.93945 – 0.95312	majority 0.36328 – 0.38672; shuffled 0.24805 – 0.29492; oracle 1 – 1	s0: 0.2793 – 0.42188; s1: 0.26953 – 0.45898; s2: 0.29102 – 0.44922	fixed budget
r10 records → prediction+sums equal	upper linear sum	s0: 0.17773 – 0.25195; s1: 0.16797 – 0.25; s2: 0.16797 – 0.23828	s0: 0.03125 – 0.052734; s1: 0.033203 – 0.064453; s2: 0.027344 – 0.0625	majority 0.36328 – 0.38672; shuffled 0.0097656 – 0.03125; oracle 0.99805 – 1	s0: -0.25977 – -0.099609; s1: -0.25 – -0.10151; s2: -0.23242 – -0.10737	all seeds
r10 records → prediction+sums equal	upper mlp64 sum	s0: 0.36719 – 0.43945; s1: 0.36133 – 0.41016; s2: 0.34961 – 0.39453	s0: 0.23242 – 0.26953; s1: 0.21875 – 0.27734; s2: 0.23633 – 0.26953	majority 0.36328 – 0.38672; shuffled 0.21094 – 0.29492; oracle 1 – 1	s0: -0.25391 – -0.048779; s1: -0.23047 – -0.044922; s2: -0.18945 – -0.0625	fixed budget
r10 records → prediction+sums original	raw linear sum	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	s0: 0.94531 – 0.96094; s1: 0.93359 – 0.96289; s2: 0.92383 – 0.95312	majority 0.36328 – 0.38672; shuffled 0.0019531 – 0.025391; oracle 0.99805 – 1	s0: 0.5644 – 0.73633; s1: 0.54883 – 0.72266; s2: 0.50586 – 0.77539	all seeds
r10 records → prediction+sums original	raw mlp64 sum	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	s0: 0.93164 – 0.95508; s1: 0.92969 – 0.96484; s2: 0.91992 – 0.95898	majority 0.36328 – 0.38672; shuffled 0.23633 – 0.28906; oracle 1 – 1	s0: 0.2812 – 0.45117; s1: 0.26953 – 0.47661; s2: 0.27148 – 0.44531	fixed budget
r10 records → prediction+sums original	upper linear sum	s0: 0.17773 – 0.25195; s1: 0.16797 – 0.25; s2: 0.16797 – 0.23828	s0: 0.29492 – 0.45898; s1: 0.26172 – 0.49023; s2: 0.32422 – 0.40234	majority 0.36328 – 0.38672; shuffled 0.0058594 – 0.029297; oracle 0.99805 – 1	s0: 0.019531 – 0.33398; s1: -0.039111 – 0.3731; s2: 0.044922 – 0.25981	all seeds
r10 records → prediction+sums original	upper mlp64 sum	s0: 0.36719 – 0.43945; s1: 0.36133 – 0.41016; s2: 0.34961 – 0.39453	s0: 0.38281 – 0.50977; s1: 0.37695 – 0.56445; s2: 0.41016 – 0.4668	majority 0.36328 – 0.38672; shuffled 0.16797 – 0.2207; oracle 1 – 1	s0: -0.095703 – 0.17974; s1: -0.080127 – 0.20898; s2: -0.019531 – 0.15039	fixed budget
r10 records → prediction equal	raw linear sum	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	s0: 0.24023 – 0.49414; s1: 0.36133 – 0.43359; s2: 0.20898 – 0.36523	majority 0.36328 – 0.38672; shuffled 0.0019531 – 0.017578; oracle 0.99805 – 1	s0: -0.14653 – 0.20312; s1: 0.015625 – 0.16992; s2: -0.14653 – 0.21094	all seeds

Continued on next page

condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
r10 records → prediction equal	raw mlp64 sum	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	s0: 0.53125 – 0.70117; s1: 0.55469 – 0.62305; s2: 0.46094 – 0.66406	majority 0.36328 – 0.38672; shuffled 0.31445 – 0.37891; oracle 1 – 1	s0: -0.09375 – 0.14653; s1: -0.10742 – 0.095703; s2: -0.15825 – 0.17192	fixed budget
r10 records → prediction equal	upper linear sum	s0: 0.17773 – 0.25195; s1: 0.16797 – 0.25; s2: 0.16797 – 0.23828	s0: 0.0019531 – 0.019531; s1: 0.0019531 – 0.0039062; s2: 0.015625 – 0.033203	majority 0.36328 – 0.38672; shuffled 0 – 0.021484; oracle 0.99805 – 1	s0: -0.27153 – -0.14258; s1: -0.28125 – -0.13477; s2: -0.25586 – -0.11328	all seeds
r10 records → prediction equal	upper mlp64 sum	s0: 0.36719 – 0.43945; s1: 0.36133 – 0.41016; s2: 0.34961 – 0.39453	s0: 0.35938 – 0.38086; s1: 0.35352 – 0.38086; s2: 0.33789 – 0.35742	majority 0.36328 – 0.38672; shuffled 0.34375 – 0.38672; oracle 1 – 1	s0: -0.12305 – 0.060547; s1: -0.095703 – 0.062549; s2: -0.097656 – 0.054688	fixed budget
r10 records → prediction original	raw linear sum	s0: 0.24805 – 0.34375; s1: 0.26172 – 0.33984; s2: 0.20312 – 0.37305	s0: 0.89062 – 0.94922; s1: 0.89453 – 0.9375; s2: 0.875 – 0.92969	majority 0.36328 – 0.38672; shuffled 0.0078125 – 0.029297; oracle 0.99805 – 1	s0: 0.54883 – 0.71094; s1: 0.54102 – 0.68364; s2: 0.45117 – 0.76758	all seeds
r10 records → prediction original	raw mlp64 sum	s0: 0.55273 – 0.60938; s1: 0.5332 – 0.61914; s2: 0.54102 – 0.60938	s0: 0.89648 – 0.95703; s1: 0.87891 – 0.94922; s2: 0.90234 – 0.92969	majority 0.36328 – 0.38672; shuffled 0.25391 – 0.3418; oracle 1 – 1	s0: 0.27148 – 0.40625; s1: 0.24609 – 0.41602; s2: 0.25391 – 0.42773	fixed budget
r10 records → prediction original	upper linear sum	s0: 0.17773 – 0.25195; s1: 0.16797 – 0.25; s2: 0.16797 – 0.23828	s0: 0.41016 – 0.51562; s1: 0.36133 – 0.65039; s2: 0.45312 – 0.52539	majority 0.36328 – 0.38672; shuffled 0.0058594 – 0.029297; oracle 0.99805 – 1	s0: 0.16406 – 0.33594; s1: 0.14258 – 0.45898; s2: 0.16797 – 0.38281	all seeds
r10 records → prediction original	upper mlp64 sum	s0: 0.36719 – 0.43945; s1: 0.36133 – 0.41016; s2: 0.34961 – 0.39453	s0: 0.47852 – 0.56445; s1: 0.49023 – 0.72461; s2: 0.5332 – 0.63672	majority 0.36328 – 0.38672; shuffled 0.16992 – 0.24805; oracle 1 – 1	s0: 0.037061 – 0.24805; s1: 0.023438 – 0.38091; s2: 0.10156 – 0.29883	fixed budget

Table S23: r11 all recorded probes. Training-only fitting; board-identity held-out scoring. All slots, both carriers and both readout families; sum36 and category3 are separate targets. Counts retain each cell’s own denominator. Intervals are saved paired board-bootstrap gain intervals, not across-seed bands. Calibration floors do not imply predictive use; linear convergence is necessary, not sufficient, for readability. MLP convergence is not assessed under its fixed training budget. Analytical reduction only; exact per-cell and per-update rows remain released. Printed sum-reader ranges retain seed identities; category readers remain in the detailed file. Gain CI envelopes span the five saved slot intervals and are not new confidence intervals. Floor ranges span seeds and slots. Detailed records: paper/data/release/appendix_E_r11_probes.json.gz (schema and SHA-256 in release/index.json).

condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction+sums (sums dose 100%) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.90625 – 0.93359; s1: 0.91211 – 0.96289; s2: 0.89062 – 0.97266	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.50781 – 0.71289; s1: 0.5019 – 0.70117; s2: 0.5293 – 0.75781	all seeds
records → prediction+sums (sums dose 100%) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.91211 – 0.92969; s1: 0.89648 – 0.97461; s2: 0.9082 – 0.97461	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.24414 – 0.4082; s1: 0.16016 – 0.41797; s2: 0.24414 – 0.46289	fixed budget
records → prediction+sums (sums dose 100%) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.73633 – 0.81445; s1: 0.75195 – 0.81836; s2: 0.72852 – 0.8457	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.32617 – 0.55469; s1: 0.33984 – 0.57622; s2: 0.34375 – 0.62695	all seeds
records → prediction+sums (sums dose 100%) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.83008 – 0.89453; s1: 0.85742 – 0.89844; s2: 0.84766 – 0.90039	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.18164 – 0.39453; s1: 0.13281 – 0.38477; s2: 0.18164 – 0.43164	fixed budget
records → prediction+sums (sums dose 100%) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.91406 – 0.94531; s1: 0.91797 – 0.95898; s2: 0.91406 – 0.97656	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.5332 – 0.7207; s1: 0.52148 – 0.71875; s2: 0.55073 – 0.76172	all seeds
records → prediction+sums (sums dose 100%) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.91602 – 0.94922; s1: 0.91406 – 0.96875; s2: 0.91016 – 0.9707	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.24995 – 0.43359; s1: 0.19722 – 0.44146; s2: 0.2519 – 0.45708	fixed budget
records → prediction+sums (sums dose 100%) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.79492 – 0.88672; s1: 0.76172 – 0.94727; s2: 0.75195 – 0.93359	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.44922 – 0.58398; s1: 0.35742 – 0.64453; s2: 0.34375 – 0.71289	all seeds
records → prediction+sums (sums dose 100%) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.8418 – 0.92188; s1: 0.84766 – 0.95508; s2: 0.82031 – 0.94531	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.19531 – 0.40625; s1: 0.12695 – 0.39648; s2: 0.17378 – 0.47852	fixed budget

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction+sums (sums dose 10%) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.90625 – 0.9375; s1: 0.9043 – 0.94727; s2: 0.88281 – 0.93945	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.52539 – 0.72852; s1: 0.50586 – 0.6875; s2: 0.50586 – 0.72461	all seeds
records → prediction+sums (sums dose 10%) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.89648 – 0.95508; s1: 0.91016 – 0.92773; s2: 0.90039 – 0.92578	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.25776 – 0.4356; s1: 0.1875 – 0.39648; s2: 0.23628 – 0.4082	fixed budget
records → prediction+sums (sums dose 10%) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.66602 – 0.81641; s1: 0.71875 – 0.82812; s2: 0.74414 – 0.81445	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.27148 – 0.52539; s1: 0.37305 – 0.52344; s2: 0.3125 – 0.59766	all seeds
records → prediction+sums (sums dose 10%) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.79102 – 0.88672; s1: 0.83203 – 0.89844; s2: 0.82227 – 0.89258	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.18555 – 0.37109; s1: 0.16992 – 0.36914; s2: 0.16602 – 0.42578	fixed budget
records → prediction+sums (sums dose 10%) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.9043 – 0.9375; s1: 0.88672 – 0.95117; s2: 0.91211 – 0.95703	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.52734 – 0.71484; s1: 0.52539 – 0.70898; s2: 0.51367 – 0.74219	all seeds
records → prediction+sums (sums dose 10%) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.90039 – 0.94141; s1: 0.88672 – 0.94141; s2: 0.91992 – 0.96094	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.26172 – 0.42578; s1: 0.20312 – 0.41021; s2: 0.23438 – 0.44727	fixed budget
records → prediction+sums (sums dose 10%) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.78125 – 0.90625; s1: 0.70117 – 0.89258; s2: 0.73047 – 0.9043	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.42969 – 0.60938; s1: 0.33008 – 0.59766; s2: 0.38672 – 0.68364	all seeds
records → prediction+sums (sums dose 10%) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.85352 – 0.92383; s1: 0.79688 – 0.89258; s2: 0.82812 – 0.91797	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.23628 – 0.40625; s1: 0.14448 – 0.33594; s2: 0.14844 – 0.44922	fixed budget
records → prediction+sums (sums dose 1%) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.85742 – 0.91797; s1: 0.85547 – 0.92578; s2: 0.87695 – 0.92383	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.48828 – 0.69727; s1: 0.46094 – 0.68164; s2: 0.47461 – 0.71094	all seeds
records → prediction+sums (sums dose 1%) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.85938 – 0.91211; s1: 0.85547 – 0.91992; s2: 0.87891 – 0.94531	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.20117 – 0.39648; s1: 0.17383 – 0.38281; s2: 0.21875 – 0.43164	fixed budget

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction+sums (sums dose 1%) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.70508 – 0.84375; s1: 0.69531 – 0.80273; s2: 0.66602 – 0.8418	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.31055 – 0.58208; s1: 0.24805 – 0.5293; s2: 0.30078 – 0.62305	all seeds
records → prediction+sums (sums dose 1%) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.81445 – 0.87891; s1: 0.7793 – 0.8418; s2: 0.83008 – 0.89648	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.16992 – 0.37695; s1: 0.078125 – 0.33203; s2: 0.14258 – 0.42969	fixed budget
records → prediction+sums (sums dose 1%) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.86133 – 0.93359; s1: 0.83594 – 0.95312; s2: 0.85742 – 0.95508	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.51562 – 0.67188; s1: 0.46484 – 0.6895; s2: 0.5019 – 0.74219	all seeds
records → prediction+sums (sums dose 1%) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.88672 – 0.9375; s1: 0.87109 – 0.95312; s2: 0.86914 – 0.95508	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.22261 – 0.41992; s1: 0.14839 – 0.38867; s2: 0.21875 – 0.44146	fixed budget
records → prediction+sums (sums dose 1%) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.73828 – 0.86133; s1: 0.69336 – 0.90039; s2: 0.69922 – 0.90625	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.41011 – 0.55469; s1: 0.31836 – 0.59766; s2: 0.33398 – 0.68359	all seeds
records → prediction+sums (sums dose 1%) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.82812 – 0.91211; s1: 0.79492 – 0.91602; s2: 0.80078 – 0.93164	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.18159 – 0.40039; s1: 0.11909 – 0.35742; s2: 0.17578 – 0.46289	fixed budget
records → prediction+sums (Erosion continuation) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.25391 – 0.45312; s1: 0.22852 – 0.35156; s2: 0.30859 – 0.40234	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.09375 – 0.14258; s1: -0.12695 – 0.14258; s2: -0.087891 – 0.12891	all seeds
records → prediction+sums (Erosion continuation) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.52344 – 0.62305; s1: 0.49023 – 0.63086; s2: 0.58203 – 0.64648	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.13281 – 0.089844; s1: -0.13867 – 0.083984; s2: -0.054688 – 0.10156	fixed budget
records → prediction+sums (Erosion continuation) equal	upper linear sum36	s0: 0.86328 – 0.9043; s1: 0.85547 – 0.9043; s2: 0.87305 – 0.91992	s0: 0.26953 – 0.4082; s1: 0.69531 – 0.875; s2: 0.66406 – 0.75195	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.66602 – -0.43159; s1: -0.23633 – 0.054688; s2: -0.30078 – -0.10352	all seeds
records → prediction+sums (Erosion continuation) equal	upper mlp64 sum36	s0: 0.85352 – 0.89258; s1: 0.84961 – 0.9043; s2: 0.85938 – 0.89453	s0: 0.51172 – 0.57812; s1: 0.77539 – 0.90234; s2: 0.77539 – 0.89062	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.42188 – -0.23047; s1: -0.13086 – 0.060596; s2: -0.15239 – 0.033203	fixed budget

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction+sums (Erosion continuation) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.13281 – 0.83203; s1: 0.21289 – 0.74219; s2: 0.14062 – 0.80273	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.14648 – 0.51367; s1: -0.20898 – 0.48633; s2: -0.26758 – 0.59961	all seeds
records → prediction+sums (Erosion continuation) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.5 – 0.90625; s1: 0.51758 – 0.81641; s2: 0.49219 – 0.86719	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.17969 – 0.39453; s1: -0.1875 – 0.25781; s2: -0.15234 – 0.35747	fixed budget
records → prediction+sums (Erosion continuation) original	upper linear sum36	s0: 0.86328 – 0.9043; s1: 0.85547 – 0.9043; s2: 0.87305 – 0.91992	s0: 0.30078 – 0.78125; s1: 0.29688 – 0.80273; s2: 0.32617 – 0.66406	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.61133 – -0.0625; s1: -0.64258 – -0.0097656; s2: -0.60352 – -0.1582	all seeds
records → prediction+sums (Erosion continuation) original	upper mlp64 sum36	s0: 0.85352 – 0.89258; s1: 0.84961 – 0.9043; s2: 0.85938 – 0.89453	s0: 0.57617 – 0.81445; s1: 0.57031 – 0.83789; s2: 0.60547 – 0.7793	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.35352 – -0.037109; s1: -0.377 – -0.0019531; s2: -0.33008 – -0.035156	fixed budget
records → prediction (Erosion continuation) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.4375 – 0.49219; s1: 0.36133 – 0.58789; s2: 0.36523 – 0.50977	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.039014 – 0.25195; s1: -0.056641 – 0.37891; s2: -0.0097656 – 0.23833	all seeds
records → prediction (Erosion continuation) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.63672 – 0.73242; s1: 0.625 – 0.73242; s2: 0.58789 – 0.73828	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.021436 – 0.22466; s1: -0.035156 – 0.21289; s2: -0.025391 – 0.16211	fixed budget
records → prediction (Erosion continuation) equal	upper linear sum36	s0: 0.86328 – 0.9043; s1: 0.85547 – 0.9043; s2: 0.87305 – 0.91992	s0: 0.41797 – 0.49023; s1: 0.36328 – 0.55273; s2: 0.31641 – 0.45117	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.50391 – -0.32031; s1: -0.57422 – -0.29102; s2: -0.63281 – -0.38862	all seeds
records → prediction (Erosion continuation) equal	upper mlp64 sum36	s0: 0.85352 – 0.89258; s1: 0.84961 – 0.9043; s2: 0.85938 – 0.89453	s0: 0.57812 – 0.70898; s1: 0.60156 – 0.6875; s2: 0.57227 – 0.6543	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.36133 – -0.13672; s1: -0.32231 – -0.1582; s2: -0.33403 – -0.19727	fixed budget
records → prediction (Erosion continuation) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.064453 – 0.97852; s1: 0.037109 – 0.96289; s2: 0.042969 – 0.98047	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.27739 – 0.65625; s1: -0.37305 – 0.69922; s2: -0.36328 – 0.76562	all seeds
records → prediction (Erosion continuation) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.33789 – 0.98633; s1: 0.36523 – 0.97852; s2: 0.38281 – 0.98633	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.33984 – 0.46875; s1: -0.3457 – 0.41406; s2: -0.28906 – 0.47266	fixed budget

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction (Erosion continuation) original	upper linear sum36	s0: 0.86328 – 0.9043; s1: 0.85547 – 0.9043; s2: 0.87305 – 0.91992	s0: 0.2793 – 0.9375; s1: 0.23633 – 0.75977; s2: 0.11328 – 0.96289	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.6543 – 0.083984; s1: -0.70117 – -0.052734; s2: -0.8418 – 0.12305	all seeds
records → prediction (Erosion continuation) original	upper mlp64 sum36	s0: 0.85352 – 0.89258; s1: 0.84961 – 0.9043; s2: 0.85938 – 0.89453	s0: 0.42578 – 0.96875; s1: 0.40039 – 0.86914; s2: 0.44531 – 0.97461	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.47271 – 0.10352; s1: -0.54106 – 0.033203; s2: -0.48828 – 0.14844	fixed budget
records → prediction (cutoff coverage) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.47656 – 0.69531; s1: 0.45898 – 0.74414; s2: 0.41602 – 0.83398	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.087891 – 0.44536; s1: 0.10151 – 0.53325; s2: 0.074219 – 0.61328	all seeds
records → prediction (cutoff coverage) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.64648 – 0.875; s1: 0.68359 – 0.83984; s2: 0.66016 – 0.91211	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.039062 – 0.29102; s1: 0.027344 – 0.32031; s2: 0.048828 – 0.31055	fixed budget
records → prediction (cutoff coverage) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.34375 – 0.64844; s1: 0.25586 – 0.59961; s2: 0.33594 – 0.72266	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.044922 – 0.40039; s1: -0.13867 – 0.36914; s2: -0.029297 – 0.43555	all seeds
records → prediction (cutoff coverage) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.55859 – 0.83984; s1: 0.45898 – 0.64844; s2: 0.53906 – 0.80078	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.046875 – 0.30278; s1: -0.18945 – 0.12109; s2: -0.058594 – 0.20312	fixed budget
records → prediction (cutoff coverage) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.060547 – 0.98633; s1: 0.058594 – 0.97461; s2: 0.046875 – 0.98438	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.29297 – 0.66406; s1: -0.33789 – 0.70898; s2: -0.35742 – 0.76763	all seeds
records → prediction (cutoff coverage) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.38086 – 0.98438; s1: 0.38672 – 0.98828; s2: 0.37695 – 0.99414	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.25781 – 0.4668; s1: -0.33984 – 0.42192; s2: -0.2793 – 0.48052	fixed budget
records → prediction (cutoff coverage) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.074219 – 0.98438; s1: 0.052734 – 0.97461; s2: 0.037109 – 0.98828	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.29297 – 0.64453; s1: -0.39263 – 0.66997; s2: -0.39062 – 0.76172	all seeds
records → prediction (cutoff coverage) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.36914 – 0.99023; s1: 0.35742 – 0.98438; s2: 0.36523 – 0.99023	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.23828 – 0.47271; s1: -0.31836 – 0.42383; s2: -0.29883 – 0.52148	fixed budget

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction (decoy coverage) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.31055 – 0.5957; s1: 0.39062 – 0.74609; s2: 0.29688 – 0.68945	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.037109 – 0.34961; s1: 0.029297 – 0.53325; s2: -0.0039062 – 0.41797	all seeds
records → prediction (decoy coverage) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.51172 – 0.80078; s1: 0.6543 – 0.84766; s2: 0.62109 – 0.83594	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.099609 – 0.28325; s1: -0.0019531 – 0.29883; s2: 0.0097656 – 0.27539	fixed budget
records → prediction (decoy coverage) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.22656 – 0.53125; s1: 0.37695 – 0.6582; s2: 0.24023 – 0.66211	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.16211 – 0.27349; s1: -0.021484 – 0.42383; s2: -0.15044 – 0.35742	all seeds
records → prediction (decoy coverage) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.46289 – 0.7207; s1: 0.54883 – 0.72266; s2: 0.49414 – 0.78125	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.14263 – 0.18945; s1: -0.10156 – 0.18755; s2: -0.14062 – 0.22661	fixed budget
records → prediction (decoy coverage) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.070312 – 0.97852; s1: 0.074219 – 0.9668; s2: 0.068359 – 0.97461	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.27148 – 0.65625; s1: 0.34375 – 0.70117; s2: -0.33789 – 0.75977	all seeds
records → prediction (decoy coverage) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.39844 – 0.98828; s1: 0.40234 – 0.98047; s2: 0.41406 – 0.99023	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.27153 – 0.4707; s1: -0.32031 – 0.41602; s2: -0.26562 – 0.47852	fixed budget
records → prediction (decoy coverage) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.060547 – 0.97266; s1: 0.070312 – 0.96094; s2: 0.082031 – 0.97852	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.29297 – 0.63281; s1: -0.377 – 0.6582; s2: -0.3418 – 0.75195	all seeds
records → prediction (decoy coverage) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.36719 – 0.98828; s1: 0.35547 – 0.98242; s2: 0.38281 – 0.99414	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.25781 – 0.4707; s1: -0.38477 – 0.42192; s2: -0.30273 – 0.52344	fixed budget
records → prediction (uniform sampling) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.26953 – 0.58203; s1: 0.44727 – 0.6582; s2: 0.36523 – 0.55664	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.017578 – 0.32227; s1: 0.019531 – 0.4082; s2: 0.056641 – 0.3125	all seeds
records → prediction (uniform sampling) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.57617 – 0.75195; s1: 0.72266 – 0.79492; s2: 0.66797 – 0.81445	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.099658 – 0.23047; s1: -0.025439 – 0.26177; s2: -0.031299 – 0.27344	fixed budget

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction (uniform sampling) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.1875 – 0.43945; s1: 0.35352 – 0.56836; s2: 0.33203 – 0.51367	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.13672 – 0.17773; s1: -0.10352 – 0.29688; s2: 0.0097656 – 0.25786	all seeds
records → prediction (uniform sampling) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.47656 – 0.63281; s1: 0.55078 – 0.63672; s2: 0.58594 – 0.74805	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.17188 – 0.13677; s1: -0.14258 – 0.12109; s2: -0.068408 – 0.19141	fixed budget
records → prediction (uniform sampling) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.0625 – 0.96875; s1: 0.054688 – 0.98242; s2: 0.070312 – 0.9668	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.28125 – 0.64648; s1: -0.32227 – 0.71489; s2: -0.29492 – 0.75	all seeds
records → prediction (uniform sampling) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.38867 – 0.98438; s1: 0.38672 – 0.99414; s2: 0.38477 – 0.97656	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.29297 – 0.4668; s1: -0.2832 – 0.42773; s2: -0.26958 – 0.46289	fixed budget
records → prediction (uniform sampling) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.048828 – 0.98242; s1: 0.042969 – 0.98242; s2: 0.054688 – 0.97266	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.32817 – 0.64258; s1: -0.38672 – 0.67773; s2: -0.33984 – 0.74805	all seeds
records → prediction (uniform sampling) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.36523 – 0.98438; s1: 0.34961 – 0.98633; s2: 0.37891 – 0.97461	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.24609 – 0.46875; s1: -0.31641 – 0.42388; s2: -0.29297 – 0.50586	fixed budget

Table S24: r12 all recorded probes. Training-only fitting; board-identity held-out scoring. All slots, both carriers and both readout families; sum36 and category3 are separate targets. Counts retain each cell’s own denominator. Intervals are saved paired board-bootstrap gain intervals, not across-seed bands. Calibration floors do not imply predictive use; linear convergence is necessary, not sufficient, for readability. MLP convergence is not assessed under its fixed training budget. Analytical reduction only; exact per-cell and per-update rows remain released. Printed sum-reader ranges retain seed identities; category readers remain in the detailed file. Gain CI envelopes span the five saved slot intervals and are not new confidence intervals. Floor ranges span seeds and slots. Detailed records: [paper/data/release/appendix_E_r12_probes.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records+sums → prediction equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.26562 – 0.45312; s1: 0.23242 – 0.5; s2: 0.23242 – 0.33008	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.023438 – 0.14258; s1: -0.12109 – 0.29492; s2: -0.12695 – 0.10547	all seeds

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condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records+sums → prediction equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.5918 – 0.69727; s1: 0.50391 – 0.66797; s2: 0.52539 – 0.59961	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.048828 – 0.18555; s1: -0.16021 – 0.13481; s2: -0.11523 – 0.0625	fixed budget
records+sums → prediction equal	upper linear sum36	s0: 0.86328 – 0.9043; s1: 0.85547 – 0.9043; s2: 0.87305 – 0.91992	s0: 0.44531 – 0.68555; s1: 0.59766 – 0.80664; s2: 0.6543 – 0.82812	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.49224 – -0.15625; s1: -0.35552 – -0.0039062; s2: -0.29883 – -0.0038574	all seeds
records+sums → prediction equal	upper mlp64 sum36	s0: 0.85352 – 0.89258; s1: 0.84961 – 0.9043; s2: 0.85938 – 0.89453	s0: 0.54648 – 0.82227; s1: 0.7793 – 0.91211; s2: 0.75781 – 0.88867	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.28711 – -0.035156; s1: -0.16992 – 0.072266; s2: -0.17974 – 0.068359	fixed budget
records+sums → prediction original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.24609 – 0.70703; s1: 0.32227 – 0.73828; s2: 0.30469 – 0.68164	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.125 – 0.39648; s1: -0.0078613 – 0.48047; s2: -0.068359 – 0.47852	all seeds
records+sums → prediction original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.56836 – 0.85547; s1: 0.56055 – 0.83594; s2: 0.53516 – 0.86914	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.054688 – 0.34375; s1: -0.091797 – 0.27734; s2: -0.12695 – 0.36333	fixed budget
records+sums → prediction original	upper linear sum36	s0: 0.86328 – 0.9043; s1: 0.85547 – 0.9043; s2: 0.87305 – 0.91992	s0: 0.52148 – 0.88086; s1: 0.48047 – 0.80273; s2: 0.45312 – 0.8125	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.43359 – 0.029297; s1: -0.43945 – -0.0097656; s2: -0.47856 – -0.017578	all seeds
records+sums → prediction original	upper mlp64 sum36	s0: 0.85352 – 0.89258; s1: 0.84961 – 0.9043; s2: 0.85938 – 0.89453	s0: 0.72461 – 0.89648; s1: 0.67969 – 0.87891; s2: 0.63281 – 0.82422	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.17969 – 0.039062; s1: -0.2188 – 0.039062; s2: -0.30273 – 0.0058594	fixed budget
records → prediction+sums equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.90625 – 0.95117; s1: 0.9043 – 0.95898; s2: 0.92188 – 0.97656	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.53711 – 0.73633; s1: 0.50391 – 0.70898; s2: 0.51953 – 0.76177	all seeds
records → prediction+sums equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.91406 – 0.95312; s1: 0.91211 – 0.96289; s2: 0.91992 – 0.97852	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.27344 – 0.42773; s1: 0.1875 – 0.40234; s2: 0.25195 – 0.46484	fixed budget
records → prediction+sums equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.76367 – 0.86328; s1: 0.71094 – 0.8418; s2: 0.74609 – 0.82422	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.33003 – 0.61328; s1: 0.28711 – 0.58398; s2: 0.36133 – 0.60938	all seeds

Continued on next page

condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction+sums equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.86133 – 0.89453; s1: 0.79688 – 0.92383; s2: 0.82422 – 0.91602	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.21289 – 0.39258; s1: 0.11719 – 0.36523; s2: 0.17773 – 0.45117	fixed budget
records → prediction+sums original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.91406 – 0.94336; s1: 0.91602 – 0.99414; s2: 0.91211 – 0.98633	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.53315 – 0.72466; s1: 0.50391 – 0.72852; s2: 0.51172 – 0.77148	all seeds
records → prediction+sums original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.89648 – 0.94531; s1: 0.91992 – 0.98828; s2: 0.91211 – 0.99219	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.23628 – 0.41797; s1: 0.18354 – 0.42383; s2: 0.23628 – 0.48047	fixed budget
records → prediction+sums original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.71875 – 0.92578; s1: 0.7793 – 0.95898; s2: 0.75391 – 0.94727	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.32812 – 0.58984; s1: 0.34375 – 0.65625; s2: 0.34375 – 0.7207	all seeds
records → prediction+sums original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.8418 – 0.91602; s1: 0.8457 – 0.96289; s2: 0.85156 – 0.97266	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: 0.19917 – 0.40044; s1: 0.14062 – 0.4043; s2: 0.18164 – 0.50391	fixed budget
records → prediction (cutoff coverage) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.38867 – 0.65039; s1: 0.44922 – 0.63672; s2: 0.4043 – 0.63477	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.0019531 – 0.4082; s1: 0.085938 – 0.42188; s2: 0.10156 – 0.35742	all seeds
records → prediction (cutoff coverage) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.58594 – 0.8418; s1: 0.63086 – 0.82227; s2: 0.61914 – 0.82422	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.027393 – 0.33013; s1: -0.021533 – 0.23828; s2: 0.015625 – 0.25	fixed budget
records → prediction (cutoff coverage) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.32812 – 0.5; s1: 0.37305 – 0.58984; s2: 0.33789 – 0.59766	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.10552 – 0.24219; s1: -0.025391 – 0.3438; s2: -0.0078125 – 0.27935	all seeds
records → prediction (cutoff coverage) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.51562 – 0.69922; s1: 0.52539 – 0.76172; s2: 0.53125 – 0.73633	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.091797 – 0.20122; s1: -0.12505 – 0.21289; s2: -0.056689 – 0.13867	fixed budget
records → prediction (cutoff coverage) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.064453 – 0.97656; s1: 0.050781 – 0.97852; s2: 0.083984 – 0.9668	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.29297 – 0.6543; s1: -0.31445 – 0.71294; s2: -0.30273 – 0.75195	all seeds

Continued on next page

condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction (cutoff coverage) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.33984 – 0.98828; s1: 0.38086 – 0.98047; s2: 0.40234 – 0.98047	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.27148 – 0.4688; s1: -0.36719 – 0.41411; s2: -0.24419 – 0.4668	fixed budget
records → prediction (cutoff coverage) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.060547 – 0.98242; s1: 0.046875 – 0.98242; s2: 0.087891 – 0.97656	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.30278 – 0.64263; s1: -0.36133 – 0.67969; s2: -0.33203 – 0.7481	all seeds
records → prediction (cutoff coverage) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.33789 – 0.99023; s1: 0.39062 – 0.98047; s2: 0.4043 – 0.98438	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.26953 – 0.47266; s1: -0.33794 – 0.41797; s2: -0.28906 – 0.51562	fixed budget
records → prediction (decoy coverage) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.44336 – 0.67773; s1: 0.37305 – 0.74805; s2: 0.375 – 0.5625	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: 0.023389 – 0.43164; s1: 0.011719 – 0.53716; s2: 0.068359 – 0.29497	all seeds
records → prediction (decoy coverage) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.5625 – 0.84766; s1: 0.63281 – 0.85742; s2: 0.66211 – 0.77539	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.048828 – 0.29102; s1: -0.023486 – 0.33599; s2: 0.054688 – 0.21289	fixed budget
records → prediction (decoy coverage) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.26758 – 0.57227; s1: 0.36133 – 0.625; s2: 0.32812 – 0.55273	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.16602 – 0.33008; s1: -0.03125 – 0.39648; s2: 0.011719 – 0.28516	all seeds
records → prediction (decoy coverage) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.40234 – 0.74219; s1: 0.51758 – 0.79688; s2: 0.51367 – 0.70898	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.20898 – 0.20708; s1: -0.13086 – 0.25977; s2: -0.046875 – 0.1582	fixed budget
records → prediction (decoy coverage) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.070312 – 0.98828; s1: 0.060547 – 0.9707; s2: 0.089844 – 0.96289	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.29492 – 0.66411; s1: -0.32812 – 0.70513; s2: -0.27739 – 0.74609	all seeds
records → prediction (decoy coverage) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.37305 – 0.98242; s1: 0.41992 – 0.98828; s2: 0.42773 – 0.98438	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.26172 – 0.46289; s1: -0.30664 – 0.42383; s2: -0.23438 – 0.47266	fixed budget
records → prediction (decoy coverage) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.087891 – 0.98438; s1: 0.085938 – 0.97656; s2: 0.097656 – 0.96875	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.29492 – 0.64648; s1: -0.35547 – 0.67188; s2: -0.30469 – 0.74219	all seeds

Continued on next page

condition / law	reader	start accuracy: slot ranges per seed	end accuracy: slot ranges per seed	floors / ceiling ranges	gain CI envelopes per seed	converged
records → prediction (decoy coverage) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.39648 – 0.98828; s1: 0.38281 – 0.98047; s2: 0.40039 – 0.97852	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.2188 – 0.47075; s1: -0.32617 – 0.41797; s2: -0.27153 – 0.50977	fixed budget
records → prediction (uniform sampling) equal	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.30664 – 0.68945; s1: 0.26953 – 0.57227; s2: 0.25781 – 0.69531	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.080078 – 0.44336; s1: -0.089844 – 0.32617; s2: -0.042969 – 0.38672	all seeds
records → prediction (uniform sampling) equal	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.58398 – 0.83984; s1: 0.65234 – 0.79883; s2: 0.61719 – 0.82812	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.025391 – 0.31641; s1: -0.002002 – 0.22461; s2: 0.011719 – 0.25	fixed budget
records → prediction (uniform sampling) equal	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.28125 – 0.58008; s1: 0.25391 – 0.53125; s2: 0.29883 – 0.68164	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.10942 – 0.33794; s1: -0.14453 – 0.26367; s2: -0.019531 – 0.35352	all seeds
records → prediction (uniform sampling) equal	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.48047 – 0.73047; s1: 0.54688 – 0.74609; s2: 0.50391 – 0.80664	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.127 – 0.2207; s1: -0.10356 – 0.21484; s2: -0.058594 – 0.20903	fixed budget
records → prediction (uniform sampling) original	raw linear sum36	s0: 0.23633 – 0.36328; s1: 0.25977 – 0.36719; s2: 0.25391 – 0.35938	s0: 0.044922 – 0.96875; s1: 0.050781 – 0.97266; s2: 0.089844 – 0.95898	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.27734 – 0.64844; s1: -0.32031 – 0.70513; s2: -0.28516 – 0.74609	all seeds
records → prediction (uniform sampling) original	raw mlp64 sum36	s0: 0.55859 – 0.62891; s1: 0.56836 – 0.69727; s2: 0.55664 – 0.64453	s0: 0.39453 – 0.98438; s1: 0.35547 – 0.98438; s2: 0.42969 – 0.98242	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.2832 – 0.46289; s1: -0.32231 – 0.41802; s2: -0.22656 – 0.4707	fixed budget
records → prediction (uniform sampling) original	upper linear sum36	s0: 0.27539 – 0.38086; s1: 0.28516 – 0.40234; s2: 0.26758 – 0.38086	s0: 0.048828 – 0.97266; s1: 0.041016 – 0.97656; s2: 0.089844 – 0.97266	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 0.99805 – 1	s0: -0.27148 – 0.63281; s1: -0.36133 – 0.67188; s2: -0.31641 – 0.74805	all seeds
records → prediction (uniform sampling) original	upper mlp64 sum36	s0: 0.54883 – 0.60156; s1: 0.54688 – 0.6875; s2: 0.51367 – 0.64062	s0: 0.375 – 0.98438; s1: 0.37109 – 0.98047; s2: 0.41992 – 0.97852	majority 0.33984 – 0.40625; shuffled 0.16016 – 0.2168; oracle 1 – 1	s0: -0.25781 – 0.46875; s1: -0.33984 – 0.41992; s2: -0.22661 – 0.50781	fixed budget

Table S25: Original fixed-pair support and discrimination. Original registered pairs at 0/5k/20k; missing support is not a successful negative. Cutoff pairs 12/13 and 23/24 with zero held-out support are retained in the source’s cutoff_missing rows. Each recorded geometry audit has 578 missing-support pair rows; repeated audits are not independent support. The supplementary panel is separate and does not replace them. Distances and rendering variation are descriptive. Exact accuracy ceiling is 1 and balanced constant/shuffled controls calibrate the discrimination scorer. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: paper/data/release/appendix_E_original_geometry.json.gz (schema and SHA-256 in release/index.json).

condition	seed	update	measured pairs	missing support	passes	loss range	distance range	rerender range
records → prediction (sampled targets)	0	0	19	578	2	0.042453 – 192.54	0.72287 – 2.0512	0.52506 – 1.2194
records → prediction (sampled targets)	0	5,000	19	578	11	0.0011955 – 10.739	0.65817 – 2.6833	0.26142 – 1.0024
records → prediction (sampled targets)	0	20,000	19	578	11	0.00064611 – 453.03	1.0615 – 2.3784	0.27542 – 1.3484
records → prediction (sampled targets)	1	0	19	578	5	0.0010599 – 15.71	0.42328 – 1.7267	0.60117 – 1.1857
records → prediction (sampled targets)	1	5,000	19	578	9	0.0014487 – 11.654	0.72044 – 2.6499	0.21274 – 1.032
records → prediction (sampled targets)	1	20,000	19	578	13	3.4093e-05 – 7.6311	0.87523 – 2.4416	0.21569 – 1.1963
records → prediction (sampled targets)	2	0	19	578	3	0.021054 – 16.532	0.47497 – 1.779	0.62218 – 1.3293
records → prediction (sampled targets)	2	5,000	19	578	14	0.00059288 – 2.2351	0.59734 – 2.4442	0.23702 – 1.063
records → prediction (sampled targets)	2	20,000	19	578	17	0.0014398 – 24.079	0.72019 – 2.5571	0.17555 – 1.0819
records → prediction (Probability targets)	0	0	19	578	2	0.042453 – 192.54	0.72287 – 2.0512	0.52506 – 1.2194
records → prediction (Probability targets)	0	5,000	19	578	8	0.001427 – 33.443	0.94097 – 2.7274	0.231 – 0.95159
records → prediction (Probability targets)	0	20,000	19	578	11	0.00045892 – 6.3656	0.79111 – 2.683	0.21161 – 1.0337
records → prediction (Probability targets)	1	0	19	578	5	0.0010599 – 15.71	0.42328 – 1.7267	0.60117 – 1.1857
records → prediction (Probability targets)	1	5,000	19	578	11	0.00082433 – 2.2988	0.57086 – 2.657	0.17303 – 0.92151
records → prediction (Probability targets)	1	20,000	19	578	14	0.00098684 – 22.8	0.60383 – 3.1234	0.15671 – 1.0942
records → prediction (Probability targets)	2	0	19	578	3	0.021054 – 16.532	0.47497 – 1.779	0.62218 – 1.3293
records → prediction (Probability targets)	2	5,000	19	578	15	0.0017531 – 34.346	0.68991 – 2.9024	0.23508 – 0.82675
records → prediction (Probability targets)	2	20,000	19	578	11	0.00054284 – 20.875	0.8542 – 2.8308	0.22686 – 1.1654

Table S26: Supplementary cutoff discrimination and geometry support. Fixed, separately bound supplementary pairs: 135/150 for 12/13 and 12/13 for 23/24. No resampling, network updates or endpoint selection. Distances are descriptive, not a proof of absence; the calibrated decision also requires convergence and log-loss separation. Exact ceiling is accuracy 1/loss 0; constant balanced floor is 0.5/ln 2. Original support gaps (578 missing rows) remain recorded in the original geometry-support table. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_E_geometry.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

condition / seed	update	pair	fit/eval pairs	correct/ total	loss nats	shuffled accuracy	decision	distance	render distance
records → prediction (sampled targets) / 0	0	12/13	135/150	202/300	0.59475	0.44	no	1.0861	0.96062
records → prediction (sampled targets) / 0	0	23/24	12/13	18/26	0.67598	0.53846	no	1.0092	0.90475
records → prediction (sampled targets) / 0	5,000	12/13	135/150	292/300	0.0843	0.42333	no	1.1806	0.89863
records → prediction (sampled targets) / 0	5,000	23/24	12/13	22/26	0.51536	0.42308	no	1.21	0.76977
records → prediction (sampled targets) / 0	20,000	12/13	135/150	295/300	0.054223	0.3	no	1.2443	0.89607
records → prediction (sampled targets) / 0	20,000	23/24	12/13	26/26	0.015663	0.57692	yes	1.9449	1.1942
records → prediction (sampled targets) / 1	0	12/13	135/150	201/300	0.62931	0.44667	no	1.1054	1.0009
records → prediction (sampled targets) / 1	0	23/24	12/13	17/26	1.6867	0.57692	no	1.0141	0.97349
records → prediction (sampled targets) / 1	5,000	12/13	135/150	293/300	0.071934	0.35667	no	1.1464	0.89542
records → prediction (sampled targets) / 1	5,000	23/24	12/13	25/26	0.15205	0.46154	no	1.6125	0.96317
records → prediction (sampled targets) / 1	20,000	12/13	135/150	298/300	0.012802	0.36	yes	1.2043	0.91475
records → prediction (sampled targets) / 1	20,000	23/24	12/13	25/26	0.042538	0.46154	no	2.115	1.2147
records → prediction (sampled targets) / 2	0	12/13	135/150	203/300	0.60968	0.46	no	1.0427	0.95249
records → prediction (sampled targets) / 2	0	23/24	12/13	15/26	2.6461	0.46154	no	0.98879	0.95525
records → prediction (sampled targets) / 2	5,000	12/13	135/150	287/300	0.11225	0.43	no	1.1066	0.90035
records → prediction (sampled targets) / 2	5,000	23/24	12/13	26/26	0.036932	0.5	yes	1.4541	0.96377
records → prediction (sampled targets) / 2	20,000	12/13	135/150	293/300	0.060747	0.34	no	1.1628	0.89782
records → prediction (sampled targets) / 2	20,000	23/24	12/13	26/26	0.030315	0.57692	yes	1.8797	1.2604

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condition / seed	update	pair	fit/eval pairs	correct/ total	loss nats	shuffled accuracy	decision	distance	rerender distance
records → prediction (Probability targets) / 0	0	12/13	135/150	202/300	0.59475	0.44	no	1.0861	0.96062
records → prediction (Probability targets) / 0	0	23/24	12/13	18/26	0.67598	0.53846	no	1.0092	0.90475
records → prediction (Probability targets) / 0	5,000	12/13	135/150	299/300	0.01723	0.32667	yes	1.201	0.86401
records → prediction (Probability targets) / 0	5,000	23/24	12/13	26/26	0.0072501	0.57692	yes	1.5666	0.82394
records → prediction (Probability targets) / 0	20,000	12/13	135/150	300/300	0.0018263	0.32333	yes	1.2133	0.91584
records → prediction (Probability targets) / 0	20,000	23/24	12/13	26/26	0.0025216	0.69231	yes	2.1171	1.0714
records → prediction (Probability targets) / 1	0	12/13	135/150	201/300	0.62931	0.44667	no	1.1054	1.0009
records → prediction (Probability targets) / 1	0	23/24	12/13	17/26	1.6867	0.57692	no	1.0141	0.97349
records → prediction (Probability targets) / 1	5,000	12/13	135/150	300/300	0.0015462	0.27	yes	1.1757	0.82182
records → prediction (Probability targets) / 1	5,000	23/24	12/13	26/26	0.0058502	0.69231	yes	1.4999	0.85882
records → prediction (Probability targets) / 1	20,000	12/13	135/150	300/300	0.0016018	0.31333	yes	1.2355	0.88544
records → prediction (Probability targets) / 1	20,000	23/24	12/13	26/26	0.0040318	0.69231	yes	1.5396	0.86933
records → prediction (Probability targets) / 2	0	12/13	135/150	203/300	0.60968	0.46	no	1.0427	0.95249
records → prediction (Probability targets) / 2	0	23/24	12/13	15/26	2.6461	0.46154	no	0.98879	0.95525
records → prediction (Probability targets) / 2	5,000	12/13	135/150	300/300	0.0020065	0.3	yes	1.1374	0.8428
records → prediction (Probability targets) / 2	5,000	23/24	12/13	26/26	0.0076894	0.42308	yes	1.4667	0.82536
records → prediction (Probability targets) / 2	20,000	12/13	135/150	300/300	0.002165	0.35667	yes	1.2681	0.91047
records → prediction (Probability targets) / 2	20,000	23/24	12/13	26/26	0.024156	0.53846	yes	1.5231	0.90794

Table S27: r13 erosion: exact starts, first updates, minima and endpoints. Complete updates 0–200 remain in the detailed file; no interpolation or selected checkpoint. Update 0 is the matched exact-A reference; the oracle is 1,555/1,555. Blocked prediction gradients form the separate decay control, excluded from active-arm movement intersections. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_D_erosion.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

condition	seed	update 0	update 1	minimum correct	final correct	final movement
AdamW	0	1,555	393	6	48	17.647
AdamW	1	1,555	5	2	13	17.475
blocked prediction gradient	1	1,555	1,555	1,555	1,555	0.21959
blocked prediction gradient	2	1,555	1,555	1,555	1,555	0.21959
AdamW	2	1,555	518	10	14	20.514
reduced-lr AdamW	0	1,555	1,554	79	333	2.5057
reduced-lr AdamW	1	1,555	1,468	261	292	2.4478
reduced-lr AdamW	2	1,555	1,553	251	343	2.5441
SGD	0	1,555	1,440	1	1	11.424
SGD	1	1,555	89	11	11	10.21
SGD	2	1,555	1,495	2	27	36.033
blocked prediction gradient	0	1,555	1,555	1,555	1,555	0.21959

Table S28: Recorded compute and reproduction measurements. Recorded measurements only; grouped worker timings are not summed as independent training time. Unknown historical processor/GPU models remain unknown. Counts and sizes are exact, continuous values use five significant digits. Hashes, commands and the minimal archive inputs are in the portable manifest; no new training or inference occurred for these assets. Analytical reduction only; exact per-cell and per-update rows remain released. Detailed records: [paper/data/release/appendix_F_compute.json.gz](#) (schema and SHA-256 in [release/index.json](#)).

study	recorded quantity	run records	seconds range	execution scope
r10	training_seconds	18	693.71 – 2330.2	endpoint examples, not summed group/study totals
r11	elapsed_group_seconds	48	971.84 – 2797	endpoint examples, not summed group/study totals
r11	training_seconds	48	702.17 – 1942	endpoint examples, not summed group/study totals
r11	training_seconds_per_update	48	0.035109 – 0.0971	endpoint examples, not summed group/study totals
r13	elapsed_seconds	12	17.231 – 22.219	endpoint examples, not summed group/study totals
r8	elapsed_seconds	18	1661 – 1895.1	endpoint examples, not summed group/study totals

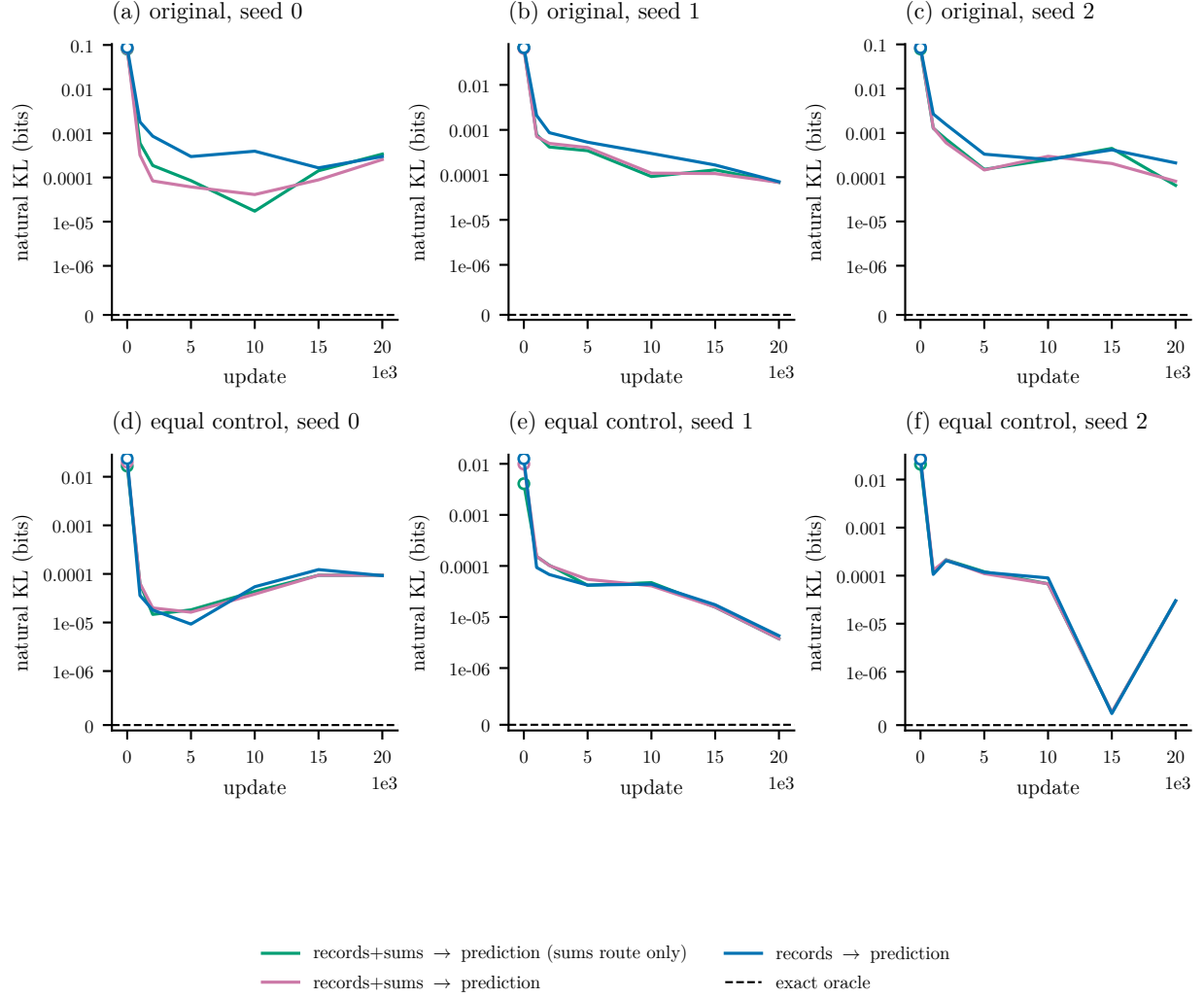


Figure S1: Recorded r8 learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

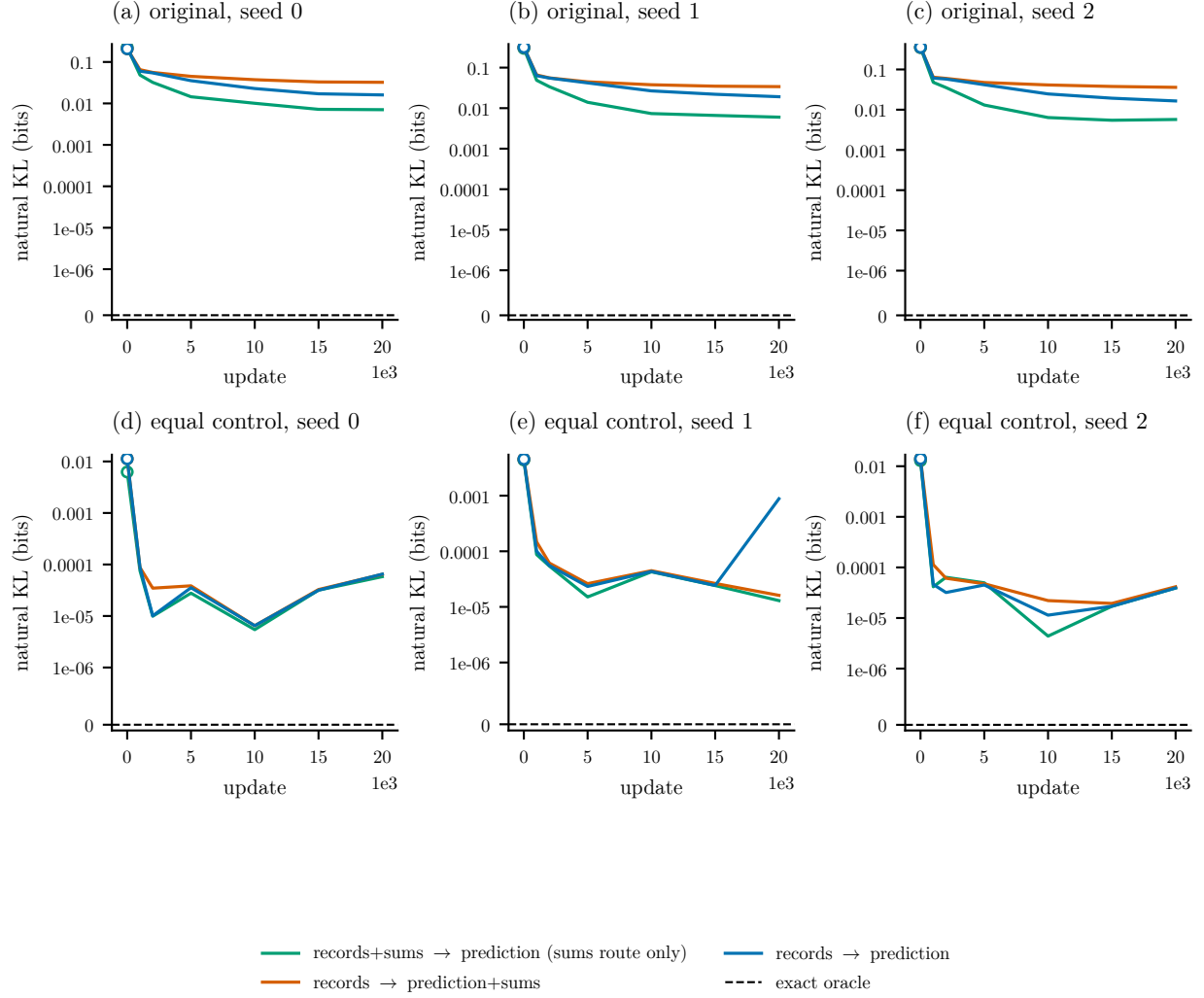


Figure S2: Recorded r10 learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

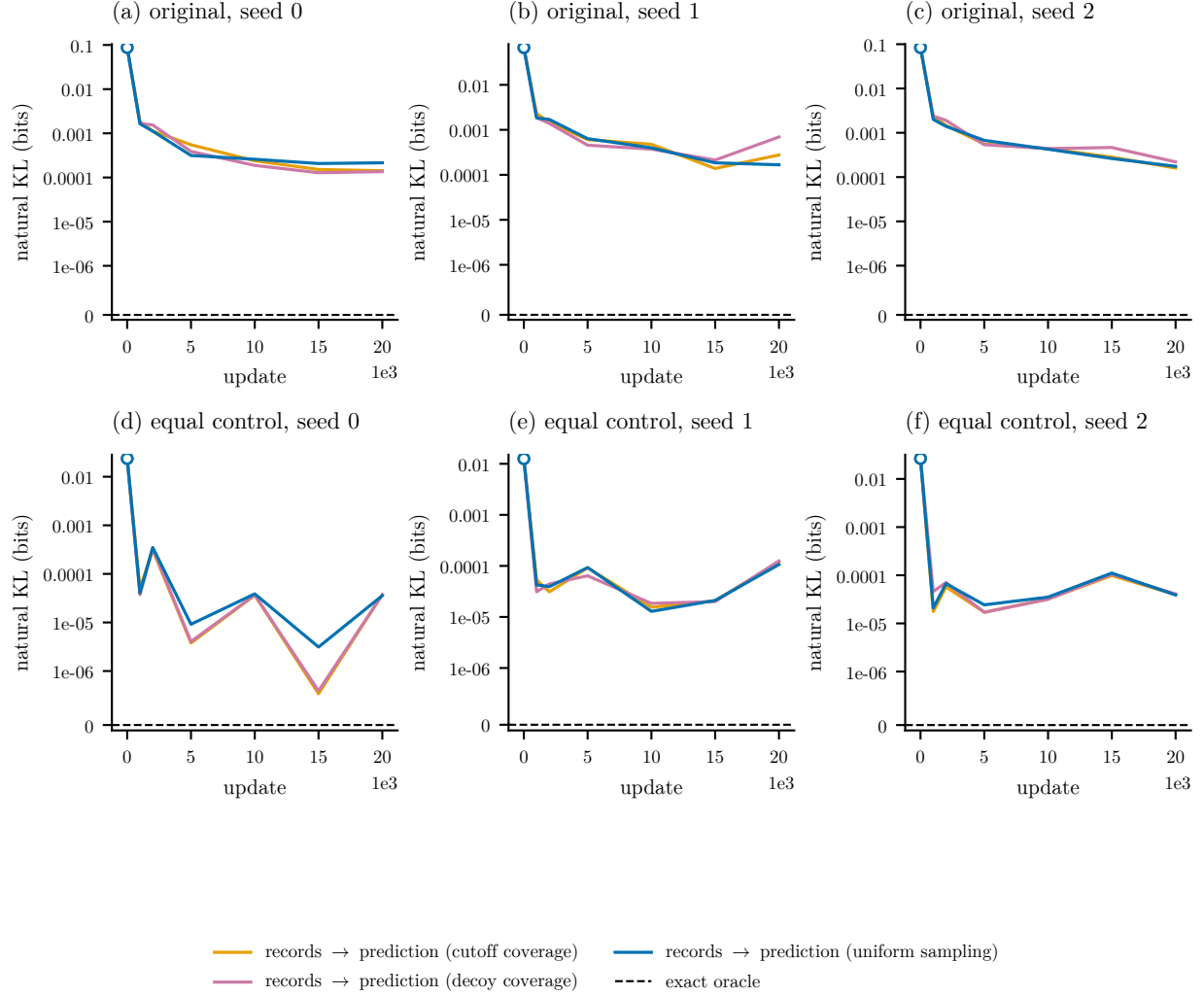


Figure S3: Recorded r11 rarity learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

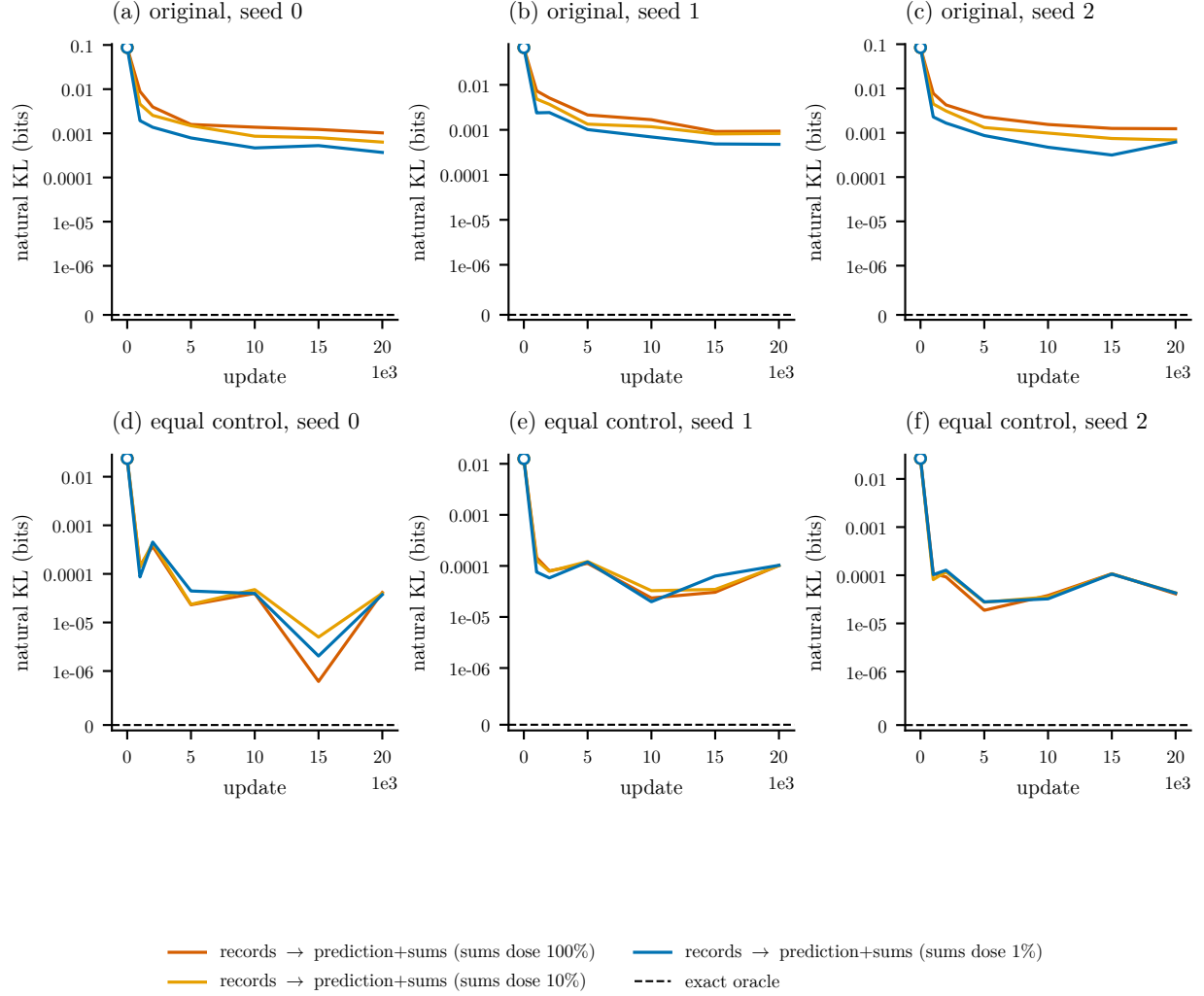


Figure S4: Recorded r11 dose learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

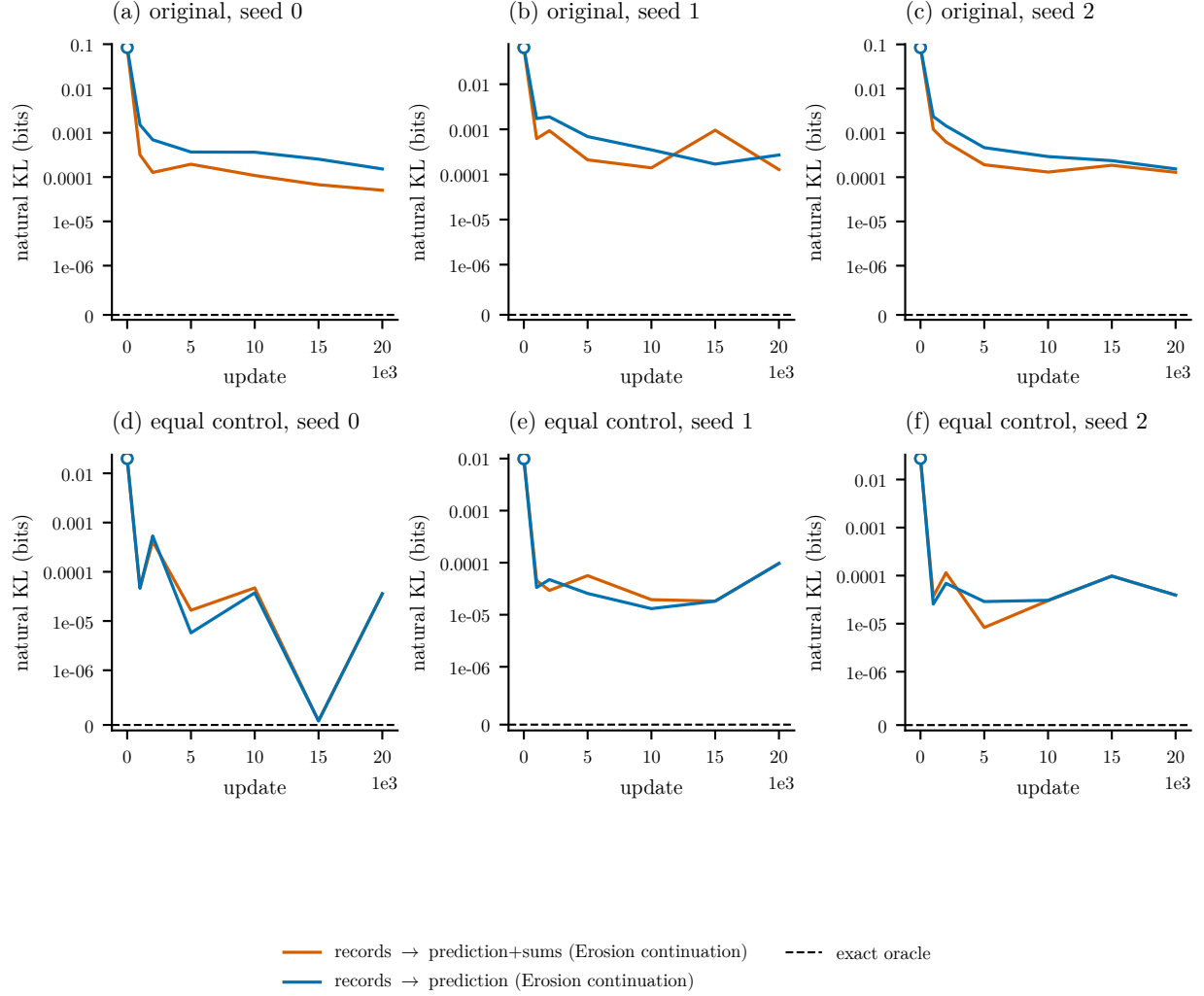


Figure S5: Recorded r11 erosion learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

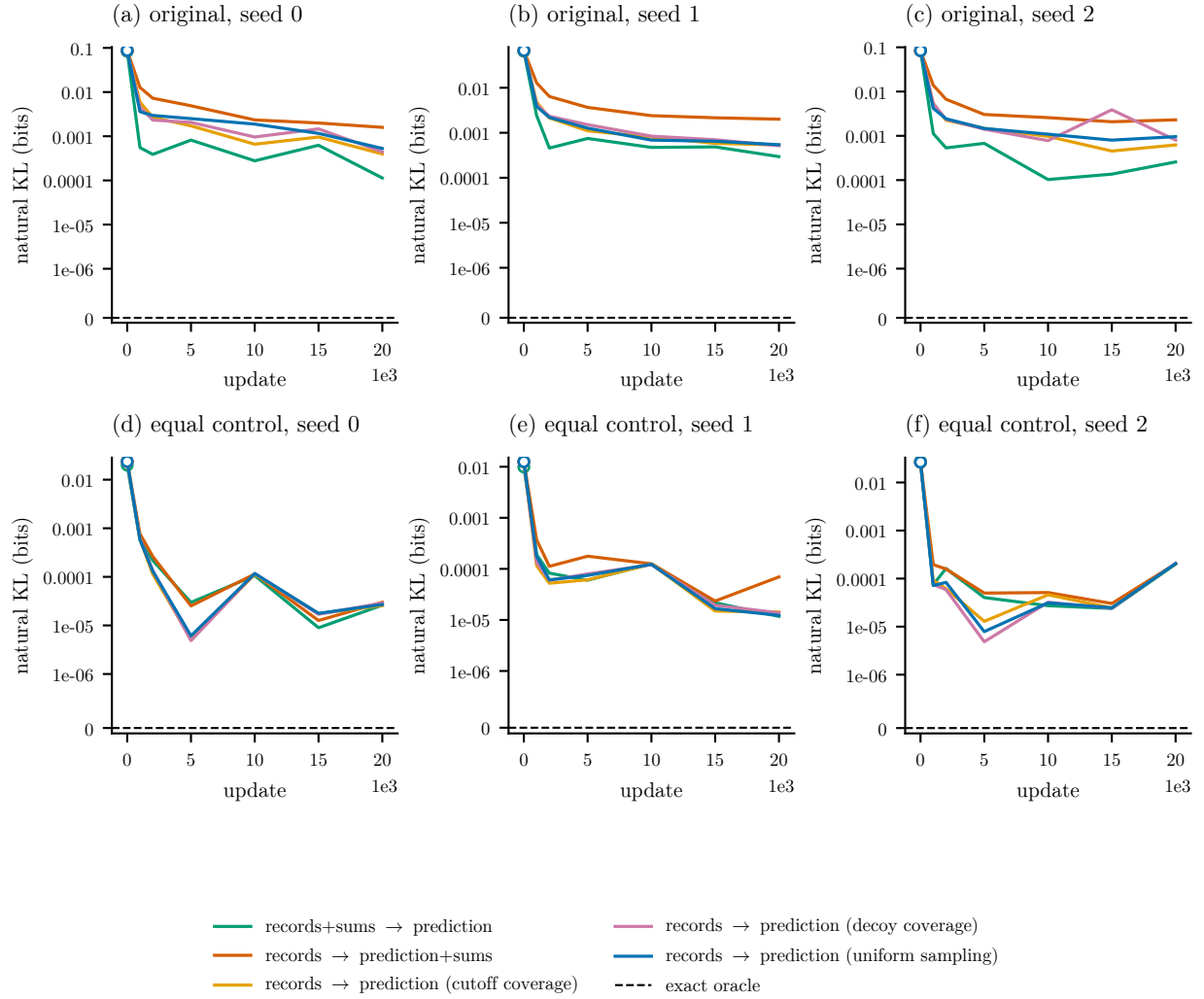


Figure S6: Recorded r12 learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

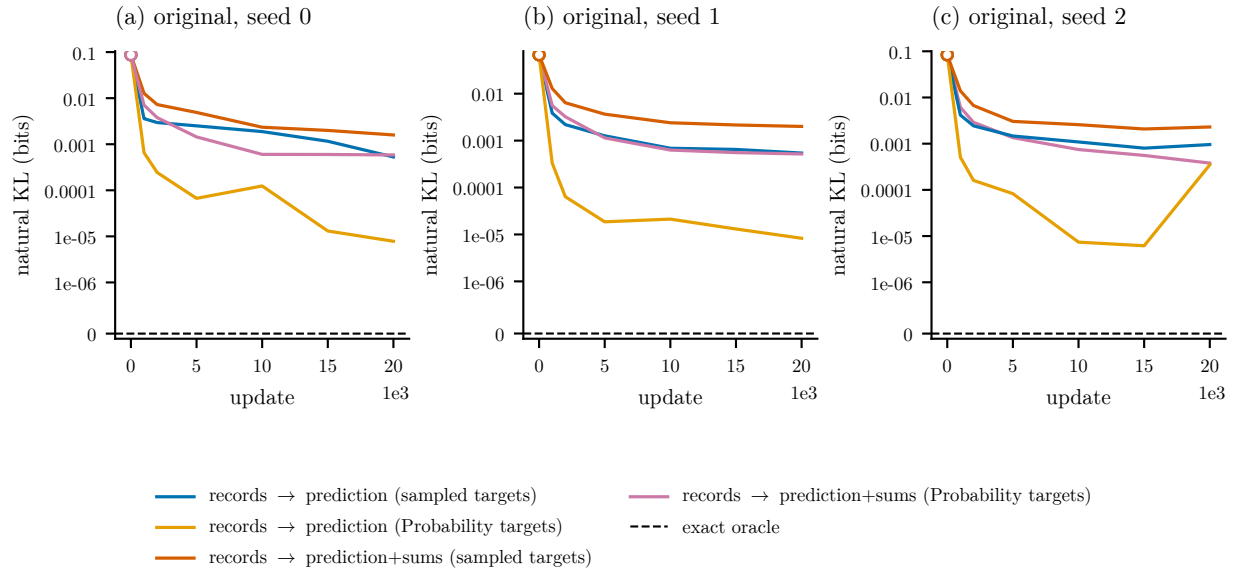


Figure S7: Recorded r13 noise learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original; no equal-law arms were run, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

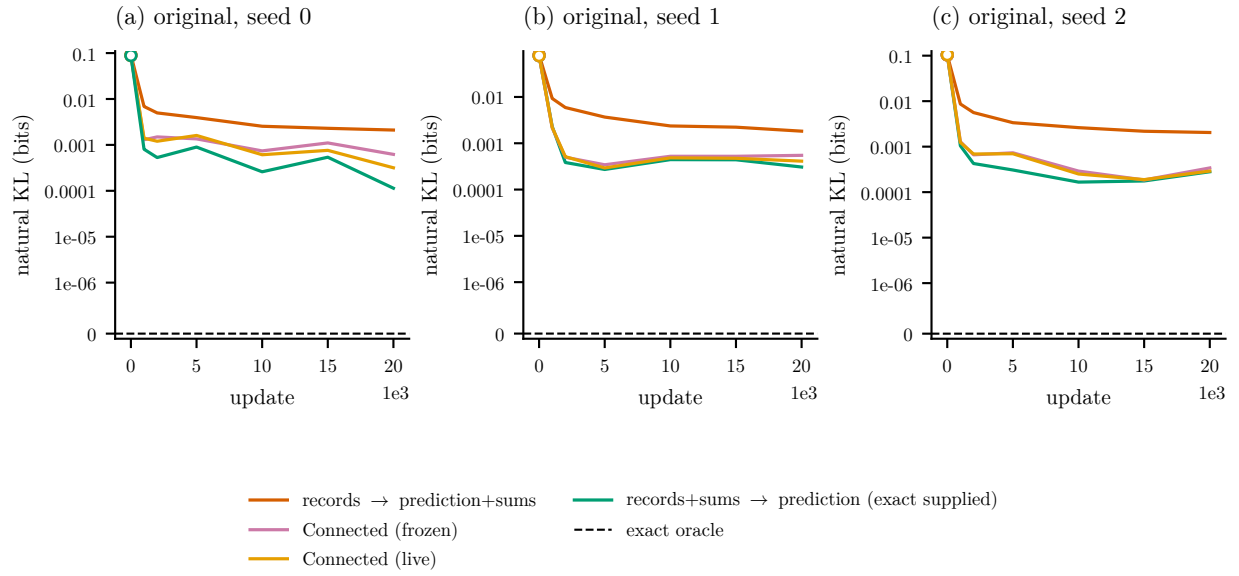


Figure S8: Recorded r13 access learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original; no equal-law arms were run, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

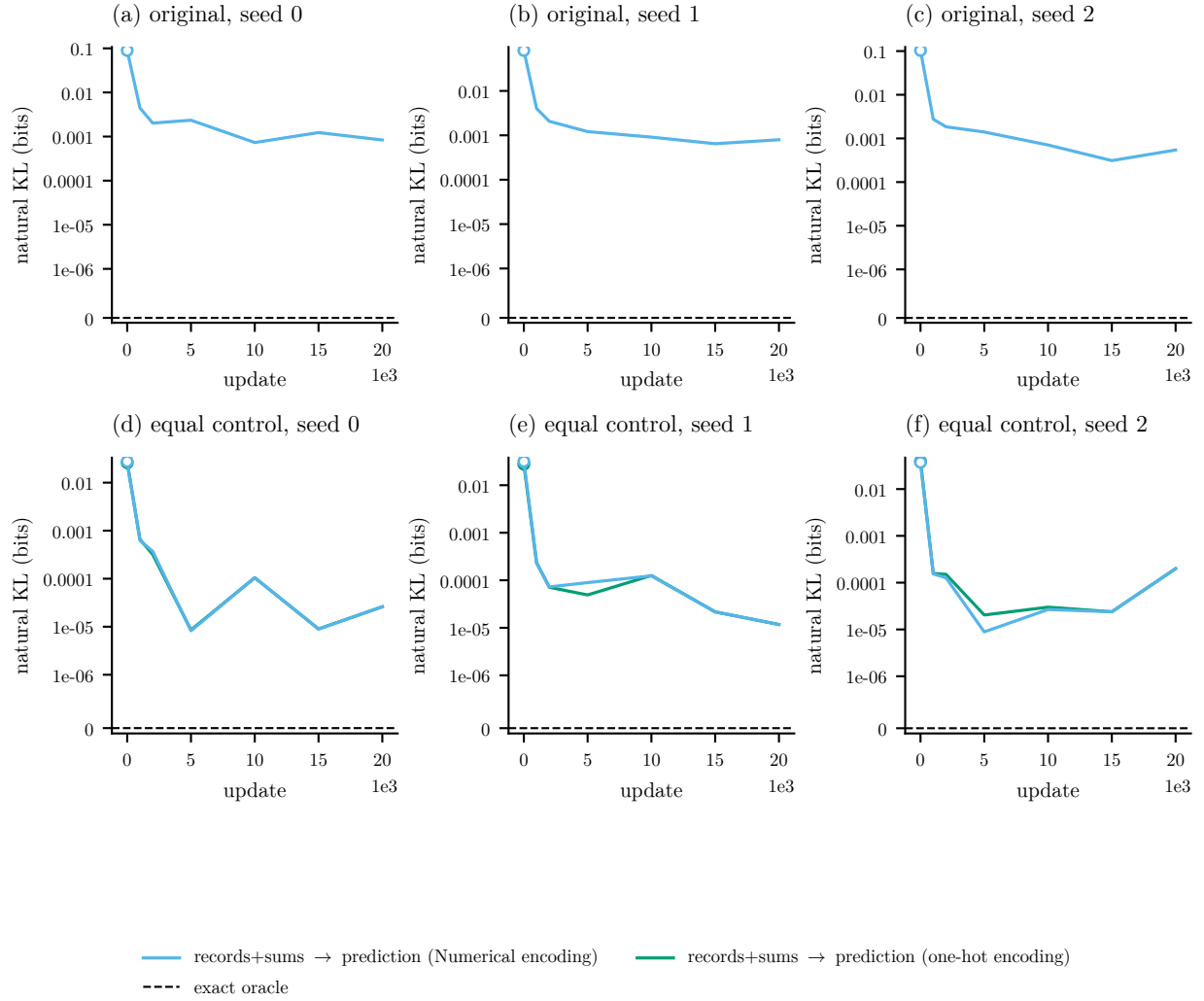


Figure S9: Recorded r13 encoding learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.